

Assignments in

BOOK 3

SENIOR MATHS

Kevin Dunne



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resources



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**SENIOR
MATHS**

BOOK **3**

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Acknowledgement

I would like to thank my wife Rikki for her invaluable assistance in the preparation of this book.

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I **introduction**

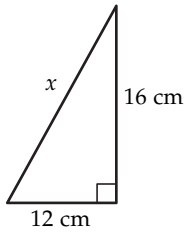
This series has been written to provide comprehensive coverage of selected topics from the senior mathematics syllabus. Book 3 concentrates on Pythagoras' theorem and trigonometry.

Most topics are aimed at the senior syllabus, but some would be useful in junior courses. Each worksheet has been carefully graded with the more difficult questions towards the end of each set. Where appropriate, exercises include a worked example to show the most common methods of working. Students should be able to complete the worksheets with minimal help from the teacher. Each topic should have sufficient questions to occupy students for at least a 40-minute lesson.

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Pythagoras' theorem can be used to find the lengths of the sides of a right-angled triangle. First you must identify the hypotenuse of the triangle.

e.g. Find the length of each unknown side, correct to one decimal place where necessary.

1

$$x^2 = 12^2 + 16^2$$

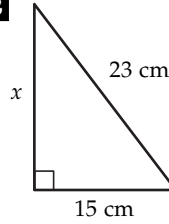
$$x^2 = 144 + 256$$

$$x^2 = 400$$

$$x = \sqrt{400}$$

$$= 20$$

$\therefore$ The length of x is 20 cm.

2

$$x^2 + 15^2 = 23^2$$

$$x^2 + 225 = 529$$

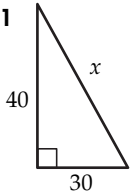
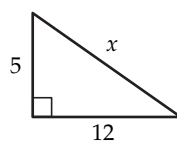
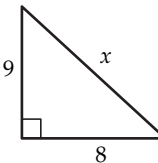
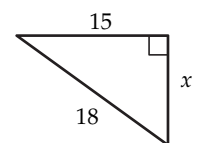
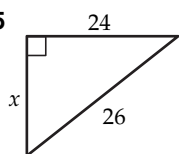
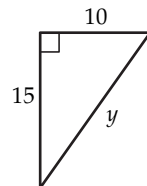
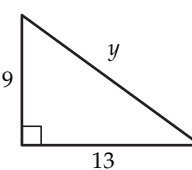
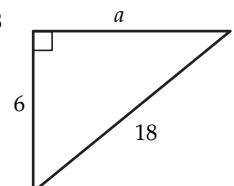
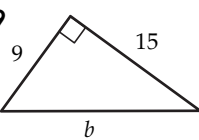
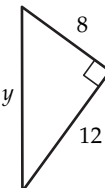
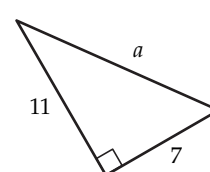
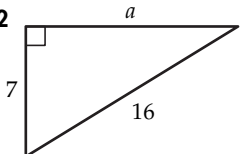
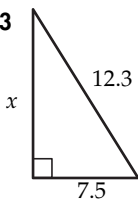
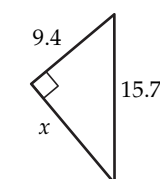
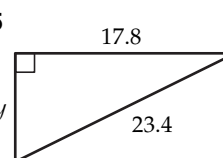
$$x^2 = 304$$

$$x = \sqrt{304}$$

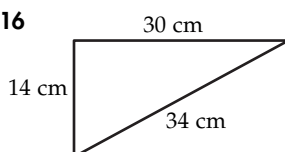
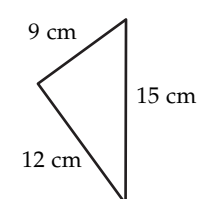
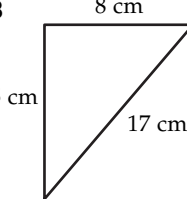
$$= 17.435\dots$$

$\therefore$ The length of x is 17.4 cm.

Find the length of each unknown side, correct to one decimal place where necessary. (All measurements are in centimetres.)

1**2****3****4****5****6****7****8****9****10****11****12****13****14****15**

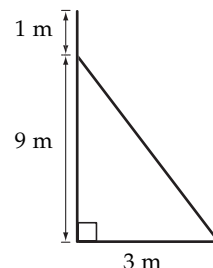
Determine whether each triangle has a right angle.

16**17****18**

19 A builder constructs a wooden frame. The lengths of the sides are 3.4 m and 6.5 m and the frame is a rectangle. Calculate the length of the diagonal to one decimal place.

20 A canoe course is in the shape of a right-angled triangle. The two legs of the course that form the right angle are 7 km and 9 km in length. The race requires each canoe to go around the course twice. What is the length of the race?

e.g. A 10-metre tower has three wire braces for support. Each brace is attached 1 metre below the top of the tower and 3 metres from its base on level ground. Determine to the nearest metre the total length of the three braces.



Let x be the length of one wire brace.

$$x^2 = 9^2 + 3^2$$

$$x^2 = 81 + 9$$

$$x^2 = 90$$

$$\therefore x = 9.487$$

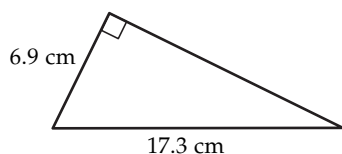
Three braces: $9.487 \times 3 = 28.461$

$\therefore$ The total length of the three braces is 28 metres.

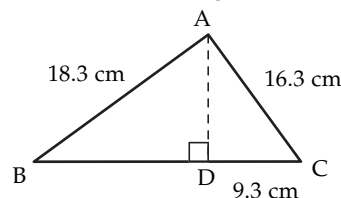
- 1 A rectangle is 18 cm long and 13 cm wide. Find the length of its diagonal.

- 2 Antonio places a 10-metre ladder against a wall. Its base is 2.5 m from the wall. How far up the wall does the ladder reach?

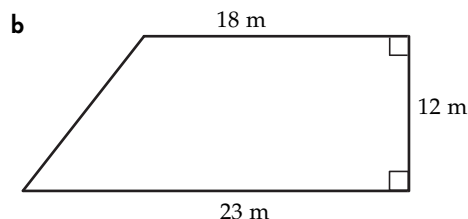
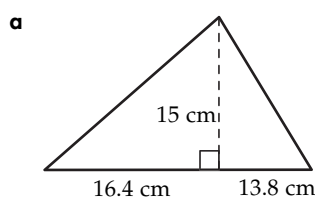
- 3 Calculate the perimeter.



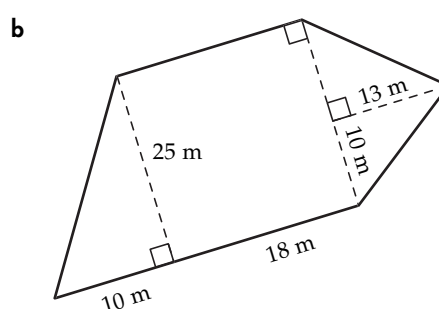
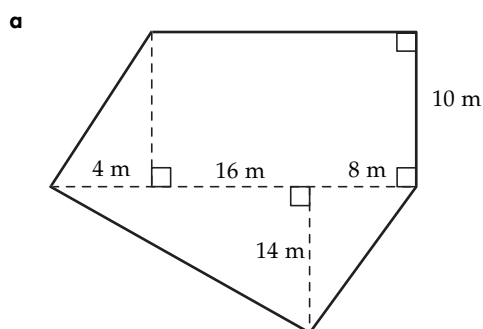
- 4 Determine the length of BD.



- 5 Determine the perimeter of each shape.



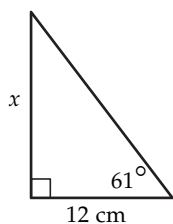
- 6 Two irregular fields were surveyed. The measurements are shown on these surveyor's sketches. Find the perimeter of each field.



- 7 Chow walks 10 km north from Town A, then walks 16 km east to Town B. Will walks directly from Town A to Town B. How much further than Will does Chow walk?
- 8 The diagonal of a square is 10.6 cm. Determine the length of its sides.
- 9 The rectangular frame of a gate is 1.8 m high and 1.4 m wide and it has one diagonal brace. Determine the length of timber needed to construct the frame.

e.g. Find the length of side x correct to one decimal place.

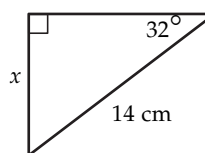
1



$$\begin{aligned}\tan 61^\circ &= \frac{x}{12} \\ x &= 12 \times \tan 61^\circ \\ &= 21.648\dots\end{aligned}$$

$\therefore$ The length of x is 21.6 cm.

2

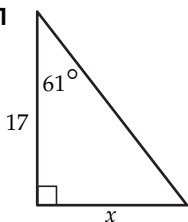


$$\begin{aligned}\sin 32^\circ &= \frac{x}{14} \\ x &= 14 \times \sin 32^\circ \\ &= 7.418\dots\end{aligned}$$

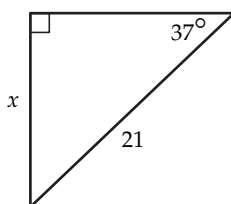
$\therefore$ The length of x is 7.4 cm.

Determine the length of side x correct to one decimal place. (All lengths are in centimetres.)

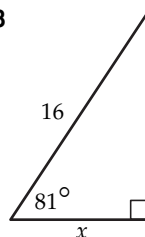
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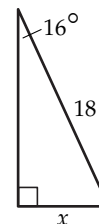
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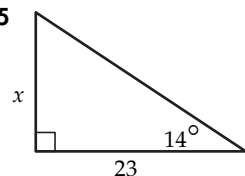
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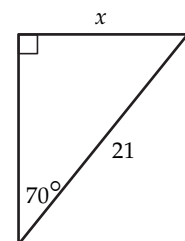
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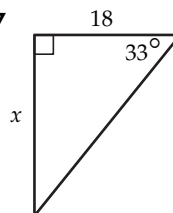
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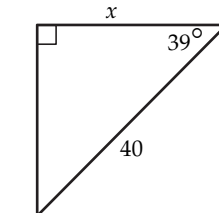
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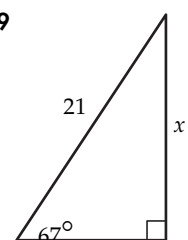
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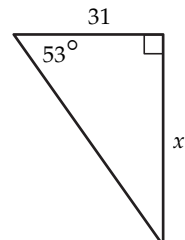
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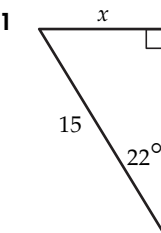
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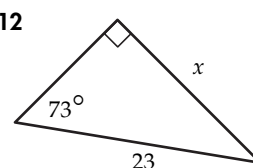
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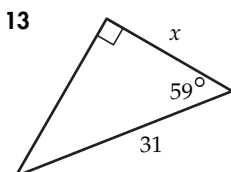
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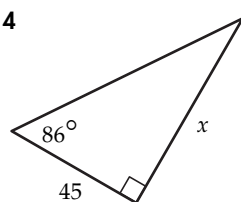
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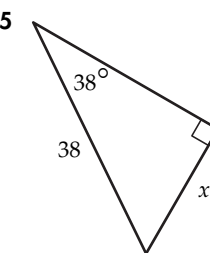
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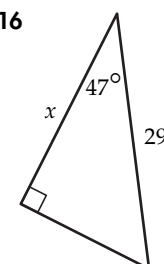
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15



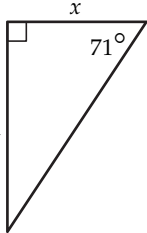
16



- 17 Triangle ABC has a right angle at C, $\angle A = 49^\circ$ and $AB = 15$ cm. Determine the length of CB to one decimal place.
- 18 $\triangle PQR$ is right-angled at R, $PR = 20$ cm and $\angle P = 38^\circ$. Find the length of QR.
- 19 In $\triangle LMN$, $\angle M = 90^\circ$, $\angle N = 57^\circ$ and $MN = 27$ cm. Find the length of LM.
- 20 In $\triangle ABC$, $\angle B = 90^\circ$, $\angle A = 64^\circ$ and $AC = 41$ cm. Calculate the length of BC.
- 21 In $\triangle XYZ$, $\angle Y = 90^\circ$, $\angle Z = 63^\circ$ and $XY = 49.6$ cm. Calculate the length of YZ.

e.g. Find the length of side x correct to one decimal place.

1



$$\tan 71^\circ = \frac{9}{x}$$

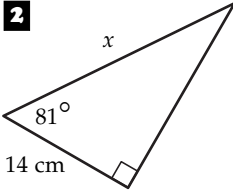
$$x \tan 71^\circ = 9$$

$$x = \frac{9}{\tan 71^\circ}$$

$$= 3.098\dots$$

$\therefore$ The length of x is 3.1 cm.

2



$$\cos 81^\circ = \frac{14}{x}$$

$$x \cos 81^\circ = 14$$

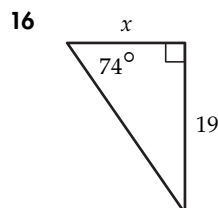
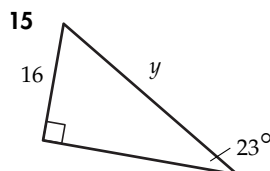
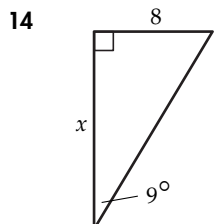
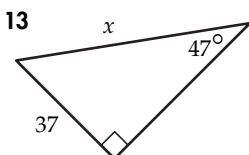
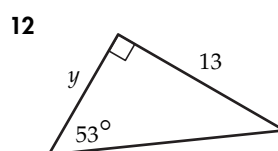
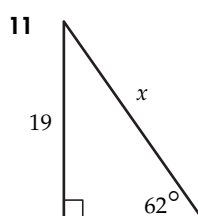
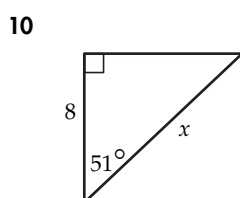
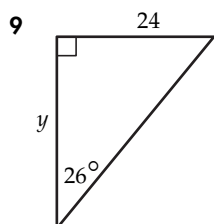
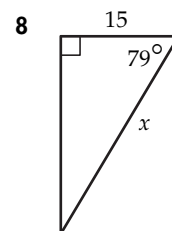
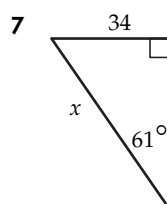
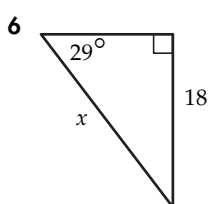
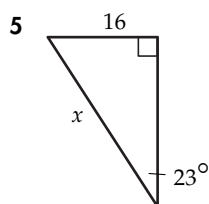
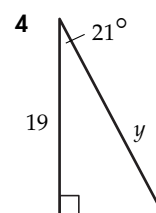
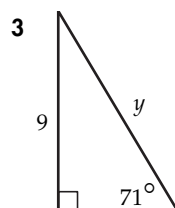
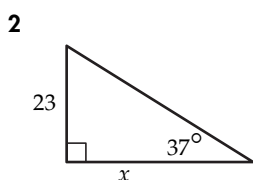
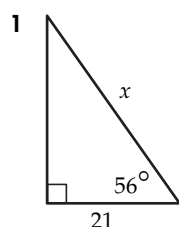
$$x = \frac{14}{\cos 81^\circ}$$

$$= 89.494\dots$$

$\therefore$ The length of x is 89.5 cm.

Note: When you need the tangent ratio and the unknown side is the denominator you can use the third angle. In Example 1, you could use $\tan 19^\circ = \frac{x}{9}$.

Determine the length of each indicated side correct to one decimal place.
(All lengths are in centimetres.)



17 In $\triangle ABC$, $\angle C$ is a right angle, $\angle B = 41^\circ$ and CA is 12 cm long. Calculate the length of AB .

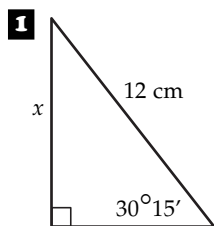
18 In $\triangle LMN$ the right angle is at N , $LN = 13$ cm and $\angle L = 25^\circ$. Find the length of LM .

19 In $\triangle PQR$, $\angle Q = 90^\circ$, $\angle P = 16^\circ$ and $QP = 21$ cm. Find the length of PR .

20 In $\triangle XYZ$, $\angle Z = 51^\circ$, $\angle Y = 90^\circ$ and $XY = 12.5$ cm. Find the length of XZ .

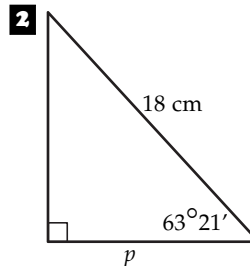
21 In $\triangle ABC$, $\angle C$ is a right angle, $\angle A = 71^\circ$ and $CB = 19.3$ cm. Determine the length of AC .

e.g. Find the length of each indicated side correct to one decimal place.



$$\begin{aligned}\sin 30^{\circ}15' &= \frac{x}{12} \\ x &= 12 \times \sin 30^{\circ}15' \\ &= 6.045\ldots\end{aligned}$$

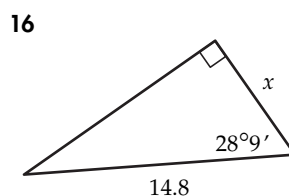
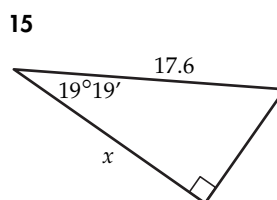
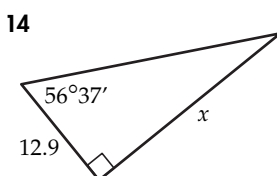
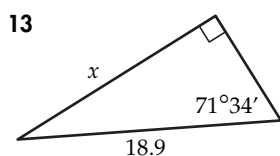
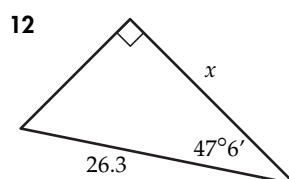
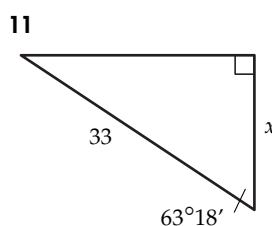
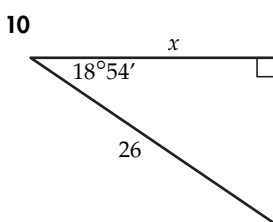
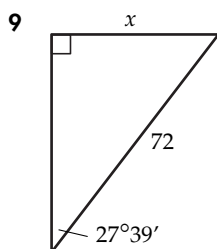
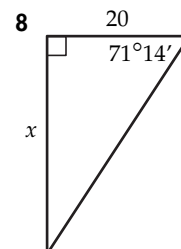
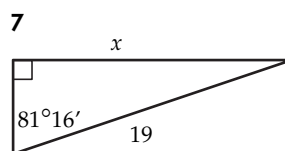
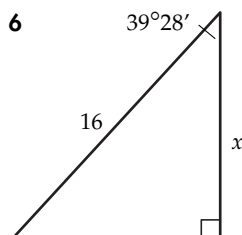
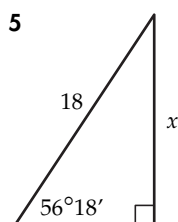
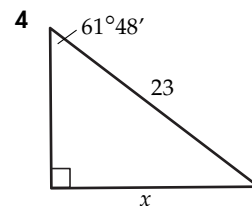
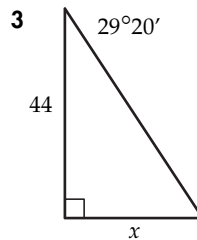
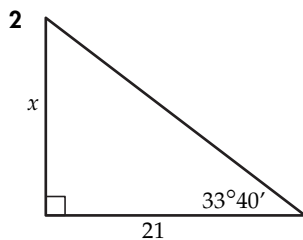
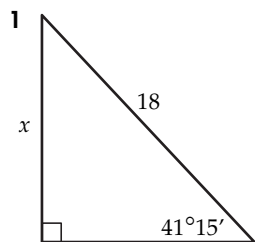
$\therefore$ The length of x is 6.0 cm.



$$\begin{aligned}\cos 63^{\circ}21' &= \frac{p}{18} \\ p &= 18 \times \cos 63^{\circ}21' \\ &= 8.073\ldots\end{aligned}$$

$\therefore$ The length of p is 8.1 cm.

Determine the length of side x correct to one decimal place. (All lengths are in centimetres.)



17 In $\triangle PQR$ the right angle is at R , $PR = 20$ cm and $\angle P = 36^{\circ}37'$. Determine the length of QR .

18 In $\triangle ABC$ the right angle is at C , $\angle A = 71^{\circ}5'$ and $AB = 26$ cm. Determine the length of CB .

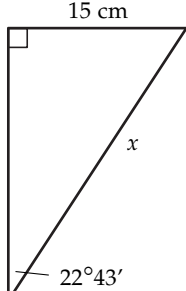
19 In $\triangle ABC$, $\angle A = 47^{\circ}15'$, $\angle B = 90^{\circ}$ and $AC = 29$ cm. Calculate the length of BC .

20 In $\triangle LMN$, $\angle N = 64^{\circ}41'$, $M = 90^{\circ}$ and $MN = 18.5$ cm. Determine the length of LM .

21 In $\triangle XYZ$, $\angle Y = 90^{\circ}$, $\angle Z = 48^{\circ}27'$ and $XZ = 46.3$ cm. Find the length of YZ .

e.g. Find the length of each side x correct to one decimal place.

1



$$\sin 22^\circ 43' = \frac{15}{x}$$

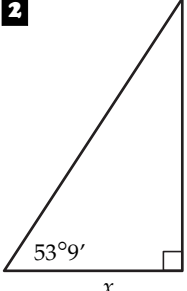
$$x \sin 22^\circ 43' = 15$$

$$x = \frac{15}{\sin 22^\circ 43'}$$

$$= 38.842\dots$$

$\therefore$ The length of x is 38.8 cm.

2



$$\tan 53^\circ 9' = \frac{5}{x}$$

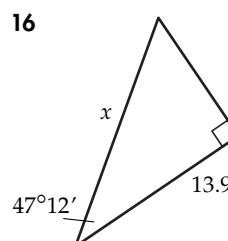
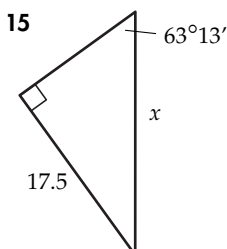
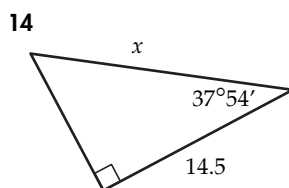
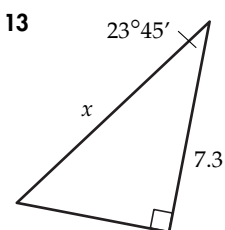
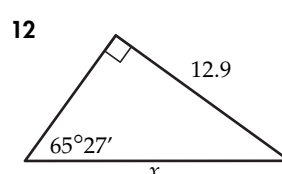
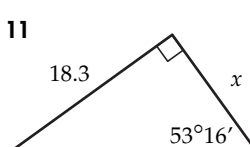
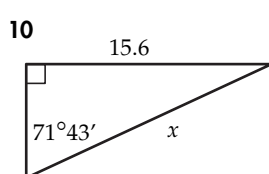
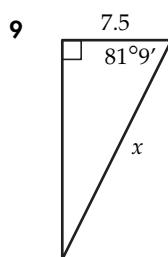
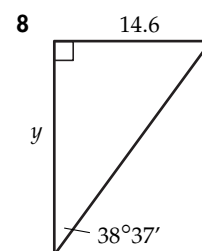
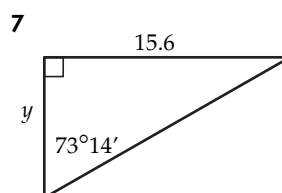
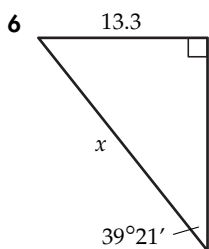
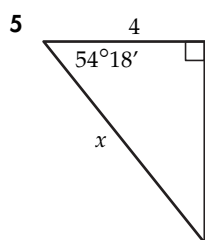
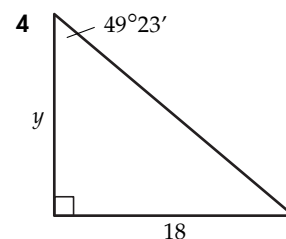
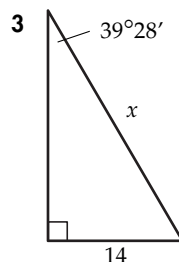
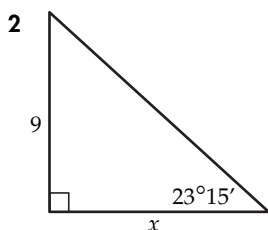
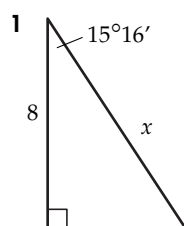
$$x \tan 53^\circ 9' = 5$$

$$x = \frac{5}{\tan 53^\circ 9'}$$

$$= 3.747\dots$$

$\therefore$ The length of x is 3.7 cm.

Determine the length of each indicated side correct to one decimal place.
(All lengths are in centimetres.)



17 In $\triangle ABC$ the right angle is at B, $\angle A = 47^\circ 35'$ and $BC = 27.3$ cm. Calculate the length of AB.

18 In $\triangle WXY$, $\angle X = 90^\circ$, $\angle Y = 16^\circ 18'$ and $WX = 12.5$ cm. Calculate the length of WY.

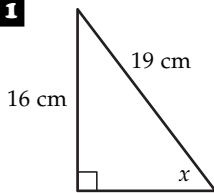
19 In $\triangle PQR$, $\angle Q = 90^\circ$, $\angle P = 73^\circ 14'$ and $QR = 19.5$ cm. Find the length of PQ.

20 $\triangle LMN$ has a right angle at M, LM has a length of 21.6 cm and $\angle L = 57^\circ 44'$. Calculate the length of LN.

21 In $\triangle DEF$, $\angle D$ is a right angle, $\angle DEF = 39^\circ 16'$ and $DF = 13.8$ cm. Find the length of the side EF.

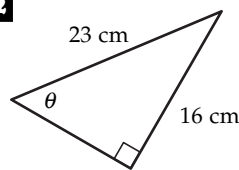
e.g. Find the indicated angle to the nearest degree.

1



$$\begin{aligned}\sin x &= \frac{16}{19} \\ \sin x &= 0.842... \\ x &= 57.363... \\ \therefore x &= 57^\circ\end{aligned}$$

2



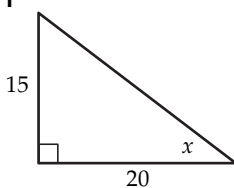
$$\begin{aligned}\cos \theta &= \frac{16}{23} \\ \cos \theta &= 0.6956... \\ \theta &= 45.92... \\ \therefore \theta &= 46^\circ\end{aligned}$$

Note: Use the or **INV** **SIN** or **2ndFN** **SIN** keys on the calculator to find the angle.

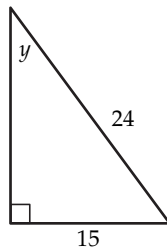
Note: Use the or **INV** **COS** or **2ndFN** **COS** keys on the calculator to find the angle.

Determine the size of each indicated angle to the nearest degree. (All lengths are in centimetres.)

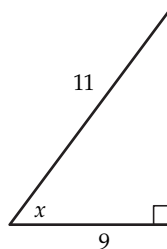
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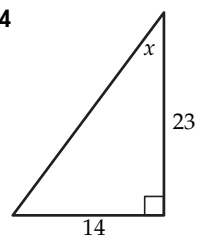
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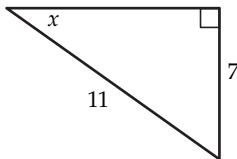
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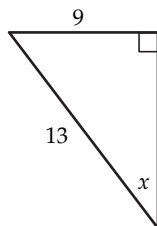
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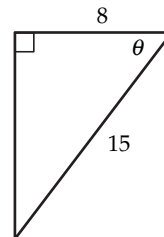
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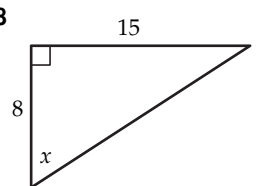
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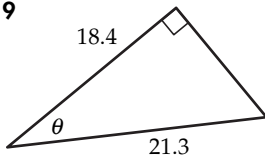
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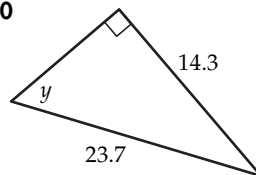
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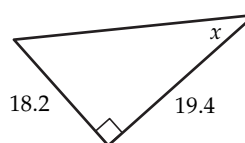
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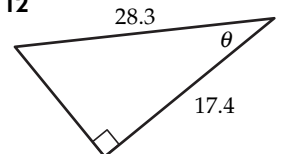
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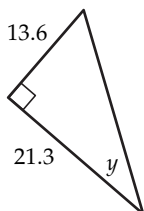
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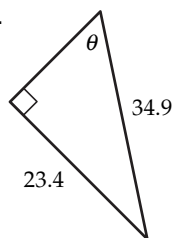
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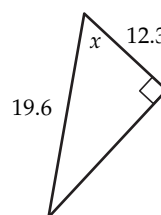
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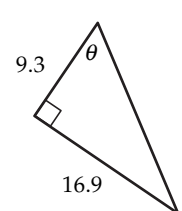
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15

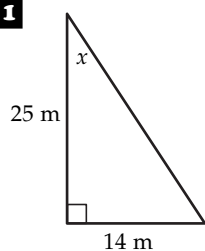


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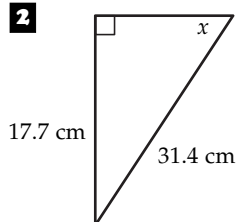


- 17** In $\triangle PQR$, $\angle Q = 90^\circ$, $PQ = 18.3$ cm and $PR = 24.6$ cm. Determine the size of $\angle R$ to the nearest degree.
- 18** In $\triangle XYZ$, $\angle Z = 90^\circ$, $ZY = 21.6$ cm and $XZ = 15.3$ cm. Find the size of $\angle X$ to the nearest degree.
- 19** $\triangle LMN$ has a right angle at N , $LN = 16.3$ cm and $MN = 14.6$ cm. Find, to the nearest degree, the size of $\angle L$.
- 20** In $\triangle ABC$, $\angle B = 90^\circ$, $AC = 12.5$ cm and $AB = 9.3$ cm. Find to the nearest degree the angle at vertex A .
- 21** In $\triangle DEF$, $\angle F = 90^\circ$, $DE = 32.1$ cm and $EF = 21.3$ cm. Find, to the nearest degree, the size of $\angle E$.

e.g. Calculate the size of angle x to the nearest minute.

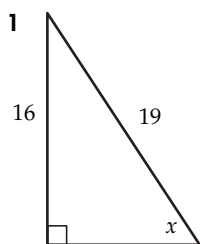
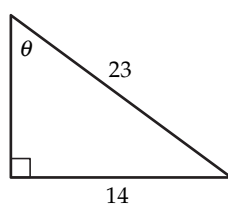
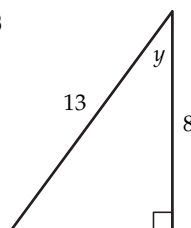
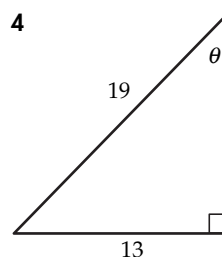
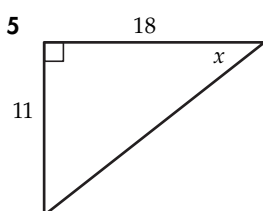
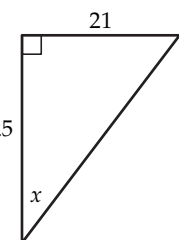
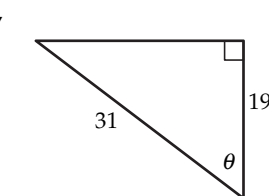
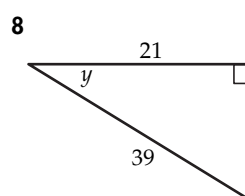
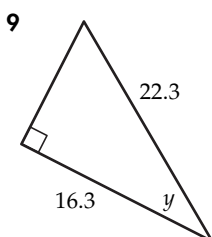
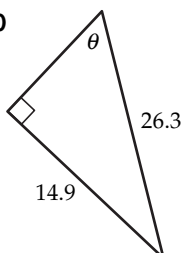
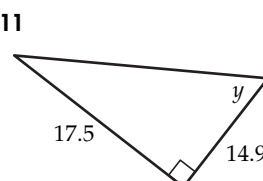
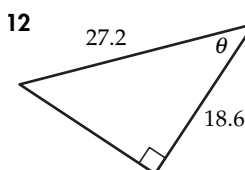
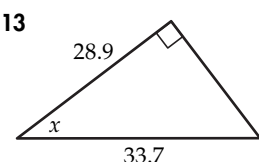
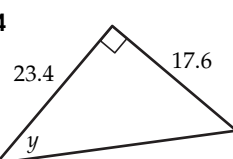
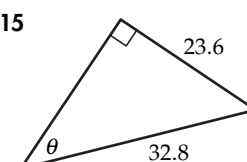
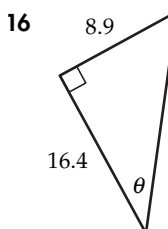
1

$$\begin{aligned}\tan x &= \frac{14}{25} \\ \tan x &= 0.56... \\ x &= 29.24... \\ \therefore x &= 29^\circ 15'\end{aligned}$$

2

$$\begin{aligned}\sin x &= \frac{17.7}{31.4} \\ \sin x &= 0.5636... \\ x &= 34.31... \\ \therefore x &= 34^\circ 19'\end{aligned}$$

Determine the size of each indicated angle to the nearest minute. (All lengths are in centimetres.)

1**2****3****4****5****6****7****8****9****10****11****12****13****14****15****16**

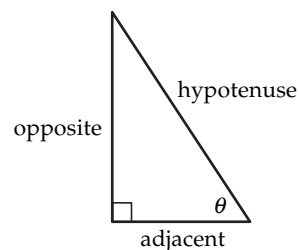
- 17** In $\triangle ABC$, $\angle B = 90^\circ$, $AC = 27.6$ cm and $AB = 18.9$ cm. Find, to the nearest minute, the size of the angle at A.
- 18** $\triangle PQR$ has a right angle at Q, $PR = 27.6$ cm and $PQ = 14.9$ cm. Calculate, to the nearest minute, the size of the angle at R.
- 19** In $\triangle LMN$, $LM = 27.9$ cm, $MN = 44.3$ cm and $\angle M = 90^\circ$. Find $\angle L$ to the nearest minute.
- 20** $\triangle CDE$ is right-angled at D, $CD = 13.9$ cm and $DE = 27.6$ cm. Determine the size of the angle at E to the nearest minute.
- 21** In $\triangle XYZ$, $XY = 16.9$ cm, $XZ = 27.6$ cm and $\angle XYZ = 90^\circ$. Find the value of $\angle XZY$ to the nearest minute.

Remember, in any right-angled triangle:

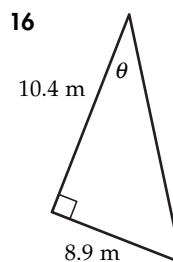
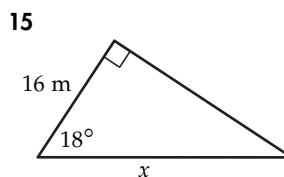
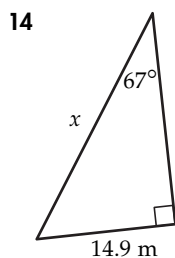
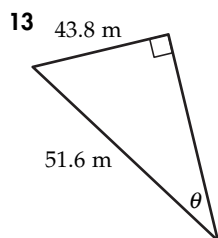
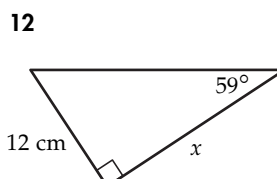
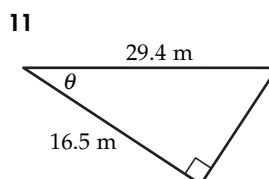
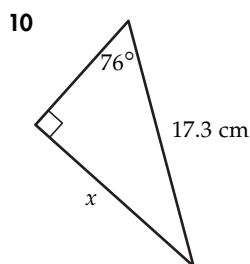
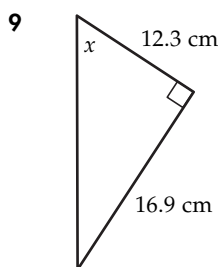
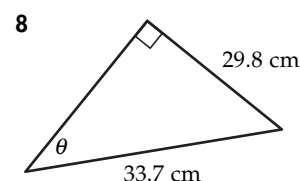
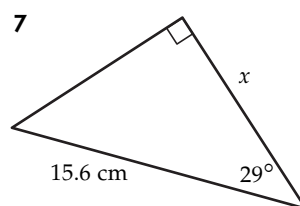
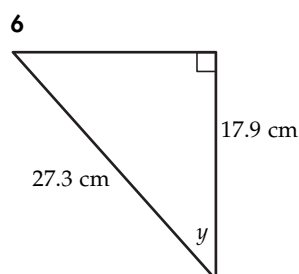
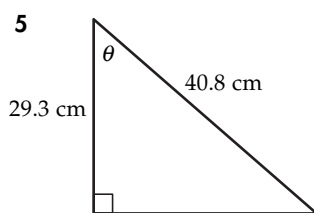
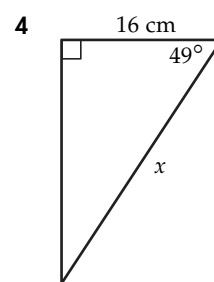
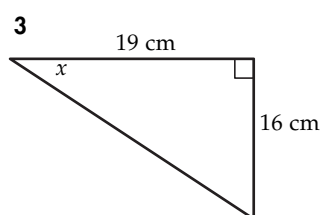
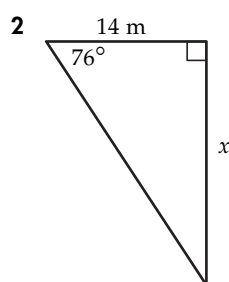
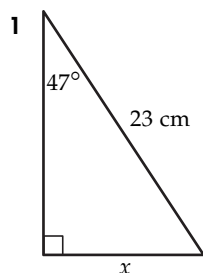
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$



Determine the value of each pronumeral. Give angles to the nearest degree and lengths correct to one decimal place.



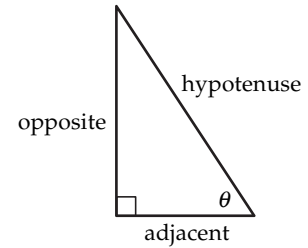
- 17 In $\triangle LMN$, $\angle M = 90^\circ$, $\angle N = 46^\circ$ and $MN = 23.4$ cm. Calculate the length of LM .
- 18 In $\triangle WST$, $\angle S = 90^\circ$, $WS = 16.9$ cm and $WT = 27.3$ cm. Find the size of $\angle WTS$.
- 19 $\triangle EFG$ has a right angle at F and sides $EF = 9.3$ cm and $FG = 6.7$ cm. Calculate the size of the angle at E .
- 20 In $\triangle WXY$, $\angle X = 90^\circ$, $\angle W = 29^\circ$ and $XY = 43.6$ cm. Find the length of WY .
- 21 $\triangle ABC$ has a right angle at C , $AB = 43.6$ cm and $AC = 33.4$ cm. Calculate the size of $\angle CAB$.

Remember, in any right-angled triangle:

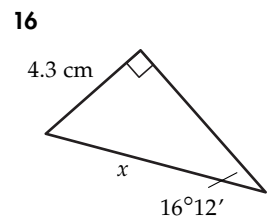
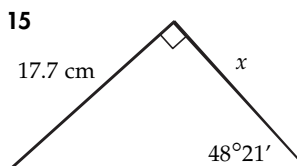
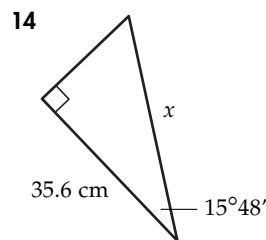
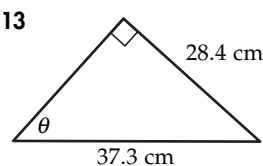
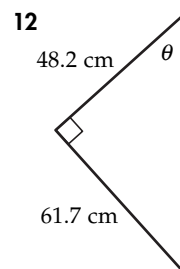
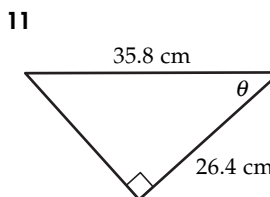
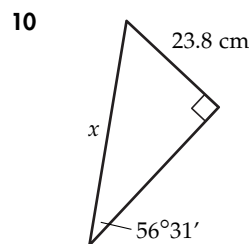
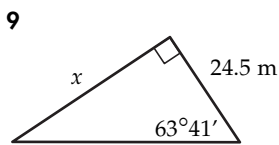
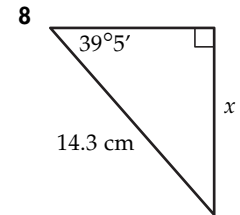
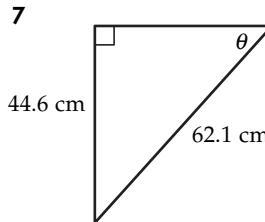
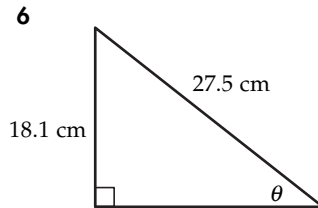
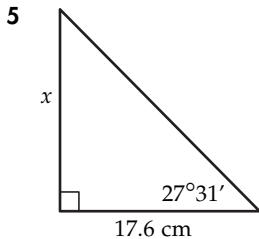
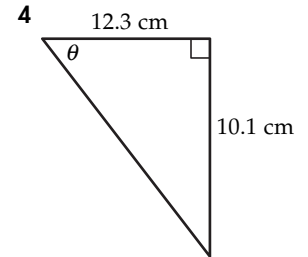
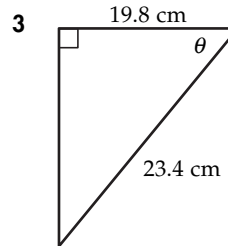
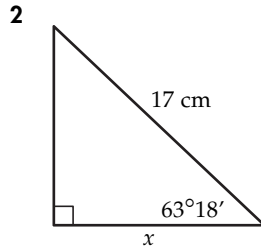
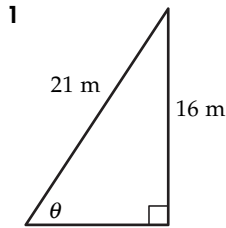
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$



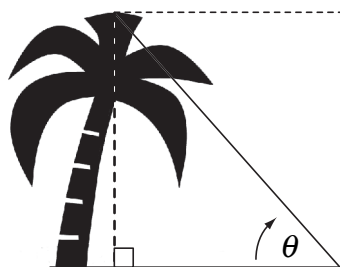
Determine the value of each pronumeral. Give angles to the nearest minute and lengths correct to one decimal place.



- 17 In $\triangle LMN$, $\angle L = 90^\circ$, $\angle N = 27^\circ 34'$ and $LM = 18.3$ cm. Determine the length of MN .
- 18 Triangle PQR is right-angled at P . Sides PQ and PR are 17.6 cm and 9.9 cm respectively. Determine the size of angle Q .
- 19 In the right-angled triangle ABC the right angle is at B , side BC is 73.6 cm and angle C is $29^\circ 3'$. Find the length of the hypotenuse.
- 20 In $\triangle XYZ$ the right angle is at Z , the length of XZ is 14.5 cm and the hypotenuse is 21.7 cm. Determine the size of the angle at X .
- 21 In $\triangle ABC$, $\angle C = 90^\circ$, $AC = 49.6$ cm and $\angle B = 64^\circ 16'$. Calculate the length of BC .

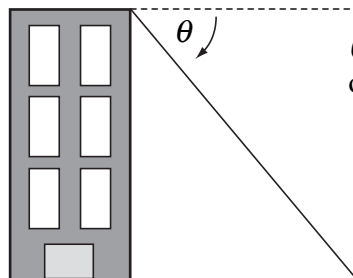
11 angles of elevation and depression

The **angle of elevation** is measured between the horizontal line and a line of sight above it.



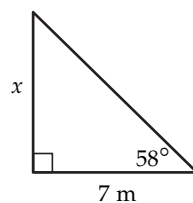
θ is the angle of elevation.

The **angle of depression** is measured between the horizontal line and a line of sight below it.



θ is the angle of depression.

e.g. Henry is calculating the height of a tree. When he stands at a point 7 metres from the base of the tree the angle of elevation is 58° . Determine the height of the tree to the nearest metre.



$$\tan 58^\circ = \frac{x}{7}$$

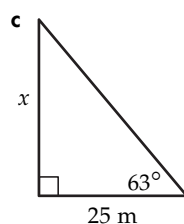
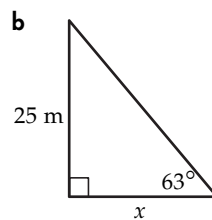
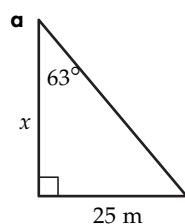
$$x = 7 \times \tan 58^\circ$$

$$x = 11.2$$

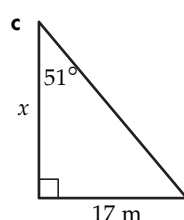
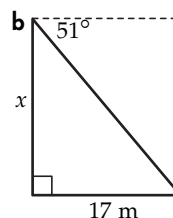
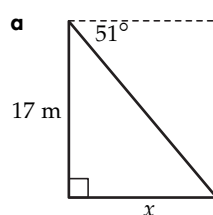
The tree is approximately 11 metres high.

For each question, identify the correct diagram and then find the value of the pronumeral.

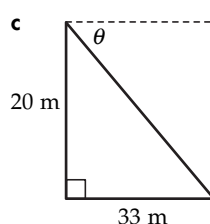
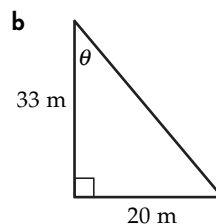
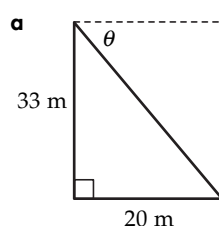
- 1 At a point 25 metres from the base of a building, the angle of elevation to the top of the building is 63° . Determine the height of the building to the nearest metre.



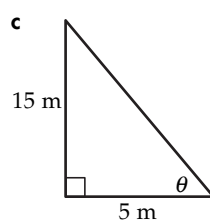
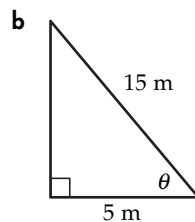
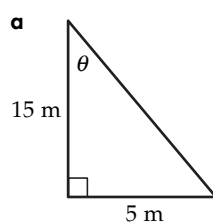
- 2 The angle of depression from the top of a building to a person standing 17 m from the base of the building is 51° . Find the height of the building to the nearest metre.



- 3 Joe is standing on the top of a building that is 33 metres high. He sees a seat in the park. The seat is 20 metres from the base of the building. Find the angle of depression to the nearest minute.



- 4 A steel brace is used to support a brick wall. The brace is 15 metres long and is attached to the ground 5 metres from the base of the wall. Determine the angle of elevation of the brace from the ground to the nearest minute.

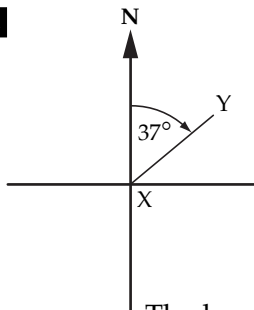


For each question, first draw and label a diagram to show the problem and then solve it.

- 5 The angle of depression from the top of a building 10 metres high to a parked car is 25° . How far (to the nearest metre) is the car from the base of the building?
- 6 Mai stands at a point on the ground that is 8 metres from the base of a tower. The tower is 25 metres high. Determine the angle of elevation to the nearest minute from where Mai stands to the top of the tower.
- 7 The angle of depression to a yacht from the top of a 20-metre high cliff is 7° . Determine the distance between the yacht and the base of the cliff to one decimal place.
- 8 Cate stands 9 metres from the base of a tree. From this point, the angle of elevation to the top of the tree is 59° . What is the height of the tree to the nearest metre?
- 9 A parked car is 31 metres from the base of a 25-metre high building. Determine the angle of depression to the nearest minute from the top of the building to the car.
- 10 A fishing boat is working in the water near a 40-metre high cliff. The angle of depression from the top of the cliff to the boat is 8° . How far (to the nearest metre) is the boat from the base of the cliff?
- 11 A wire brace is attached to a tower at a point 10.3 m above the ground. The angle of elevation of the wire is 31° . Determine the length of the wire to the nearest metre.
- 12 A wire is connected from the ground to the top of a 12.5-metre tower. The wire is 27 metres in length. Determine the angle of elevation of the wire to the tower.
- 13 The angle of elevation of a wire brace from the ground to a tower is 61° . The wire is 36.3 m. Determine the distance (to one decimal place) from the base of the tower to the point where the brace is anchored to the ground.
- 14 The angle of depression from the top of a tower along a supporting brace is 81° . The brace is anchored at a point on the ground 9.3 m from the base of the tower. Calculate the length (to one decimal place) of the brace.
- 15 A wire is stretched from the top of a 40-metre high tower to the ground. The angle of elevation of the wire to the top of the tower is 31° . Determine the length of the wire to one decimal place.
- 16 A tower has a stay cable stretched from a point 5 metres below the top of the tower to a point on the ground that is 4 metres from the tower's base. The angle of depression made by the cable is 60° . Find the height of the tower to one decimal place.
- 17 Omar is standing on the ground. He measures the angle of elevation to the top of a 150-metre tower. This angle is 87° . Determine the distance (to the nearest metre) between Omar and the base of the tower.
- 18 Susan is standing on the top of a building that is 55 m high. She notices a seat on the ground that is 23 m from the entrance to the building. Calculate, to the nearest degree, the angle of depression from the top of the building to the seat.
- 19 A rock is on the ground near the base of a tree. The distance from the rock to the tree is equal to half the height of the tree. Determine the angle of elevation (to the nearest degree) from the rock to the tree.
- 20 Freddie is standing on the ground 10 m from the base of a vertical tower. He notes that the angle of elevation to a red light on the tower is 65° and to a blue light on the tower is 54° . Calculate the distance between the two lights to one decimal place.

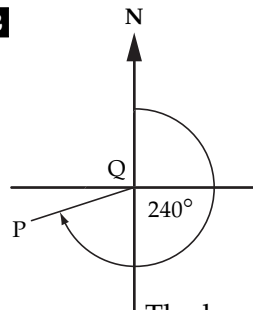
A compass bearing is a three-digit number representing the number of degrees between two points in a **clockwise** direction from **north (N)**.

e.g. 1



The bearing of Point Y from Point X is 037° .

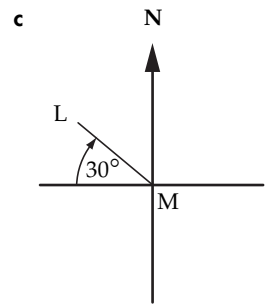
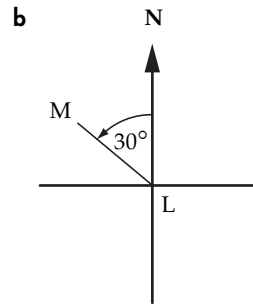
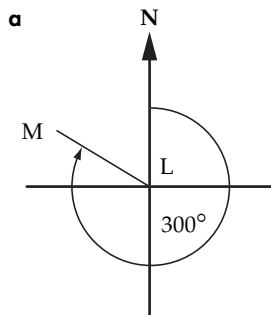
2



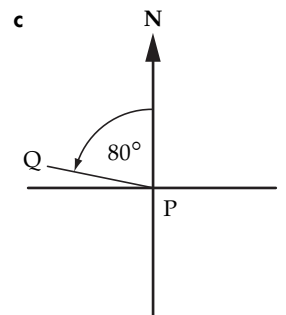
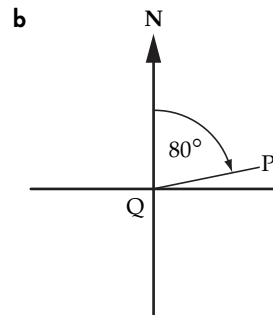
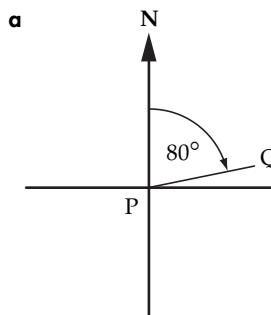
The bearing of Point P from Point Q is 240° .

Identify the diagram that best represents each bearing.

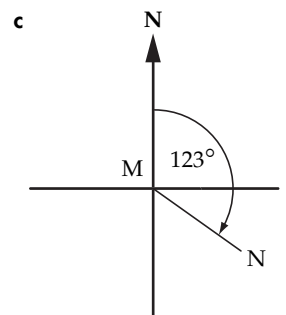
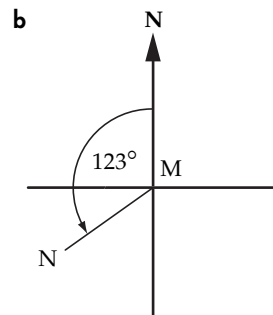
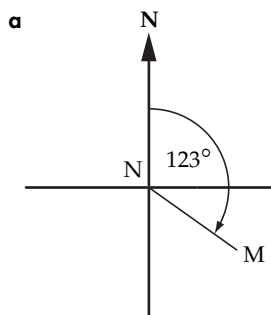
- 1 A bearing of 300° from L to M



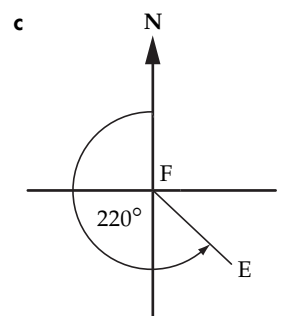
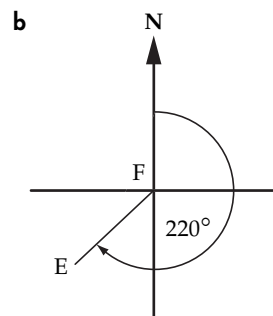
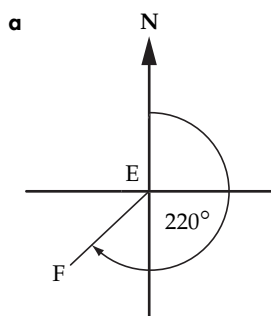
- 2 A bearing of 080° from P to Q



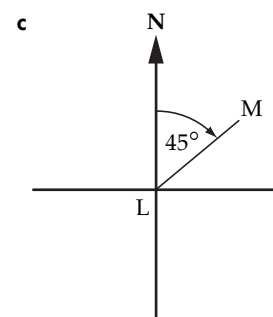
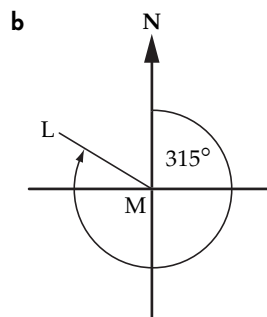
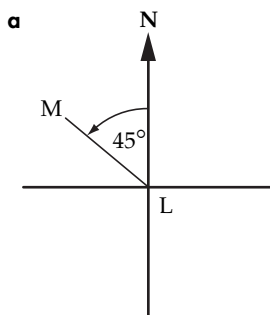
- 3 A bearing of 123° from M to N



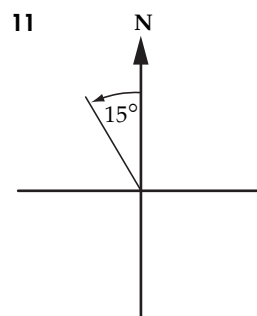
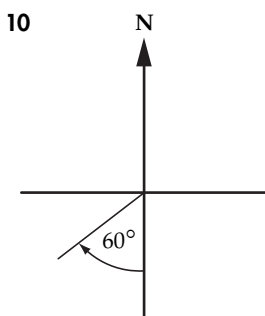
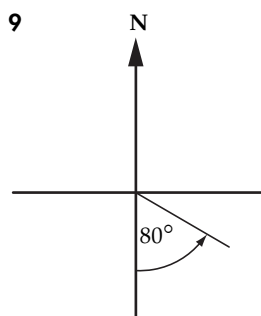
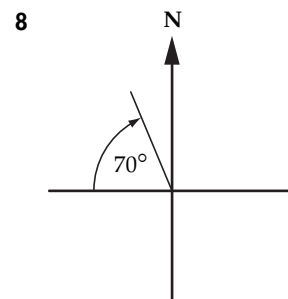
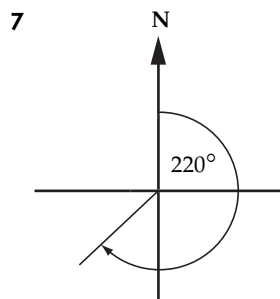
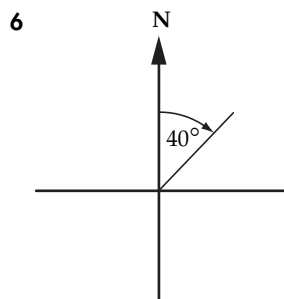
- 4 A bearing of E from F of 220°



- 5 The bearing L north west of M



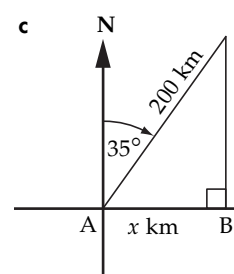
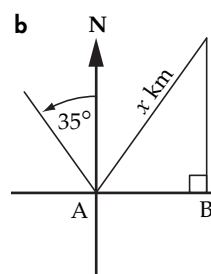
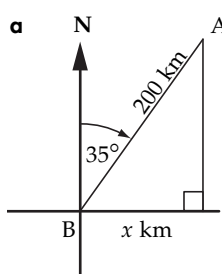
Write the bearing represented by each diagram.



Draw and label a diagram to represent each bearing.

- 12 The bearing of M from N is 110° .
 13 The bearing of O from P is 305° .
 14 The bearing of M is 010° from L.
 15 The bearing of Q is 190° from P.

- 16 A ship travels from Port A on a bearing of 035° for 200 km. Then it travels due south until it reaches Port B, which is due east of Port A. Let x represent the distance due east between Port A and the ship. Which diagram best represents this information?



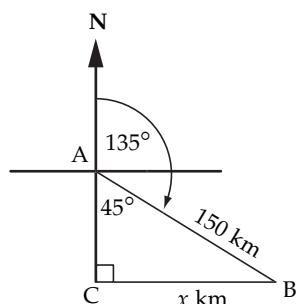
Draw neat diagrams to represent the following. Include all the given information.

- 17 Halle is in an orienteering competition. She runs for 4 km from the starting point on a bearing of 220° . Then she runs due north until she is due west of her starting point. Draw and label a diagram to represent her route, indicating with an x the line representing the distance west between the starting point and Halle.
- 18 A plane passes directly over Sydney, flying in a straight line on a bearing of 125° . It travels for a distance of 1000 km from Sydney then changes course to a bearing of 270° so that it is ultimately directly south of Sydney. Draw and label a diagram to represent the plane's route, indicating with an x the line representing the distance south between Sydney and the plane.

13 applications of compass bearings

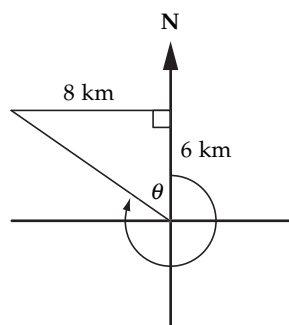
e.g. Draw a diagram of the problem, establish a right triangle and then use trigonometry to solve it.

- 1** A ship sails from Port A on a bearing of 135° for a distance of 150 km. How far east of Port A is the ship?



$$\begin{aligned}\sin 45^\circ &= \frac{x}{150} \\ x &= 150 \times \sin 45^\circ \\ x &= 106.066\dots \\ \therefore \text{The ship is } 106 \text{ km} \\ &\text{east of Port A.}\end{aligned}$$

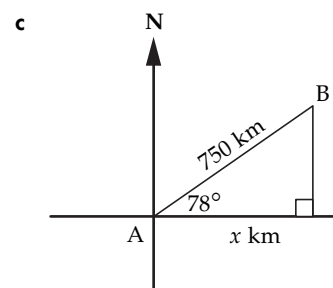
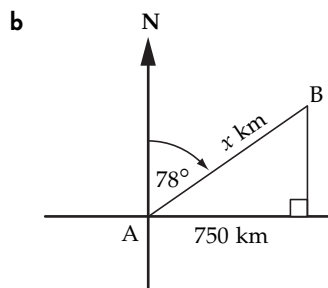
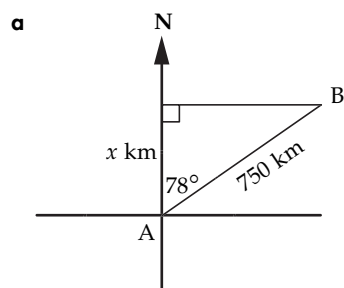
- 2** Salma leaves her house and walks due north for 6 km. Then she walks due west for a further 8 km and stops to rest. Determine the bearing of Salma from her house (to the nearest degree).



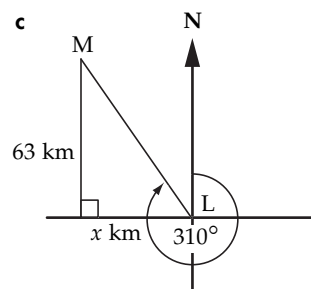
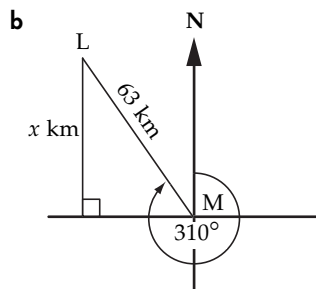
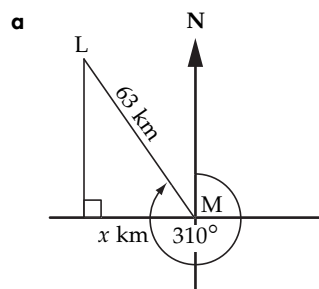
$$\begin{aligned}\tan \angle \theta &= \frac{8}{6} \\ \theta &= 53^\circ \\ \therefore \text{the bearing} &= 360^\circ - 53^\circ \\ &= 307^\circ\end{aligned}$$

For each question, identify the correct diagram and then solve the problem.

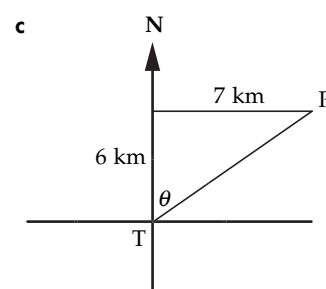
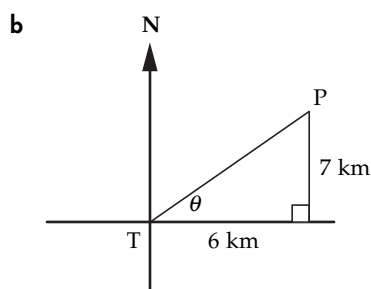
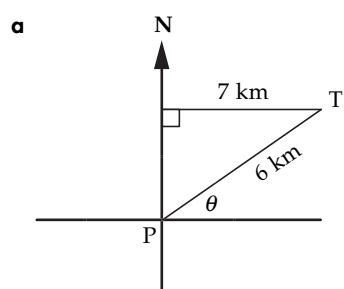
- 1** A ship sails from Port A on a bearing of 078° to Port B. The distance from Port A to Port B is 750 km. Determine how far north Port B is of Port A (that is, the value of x to the nearest km).



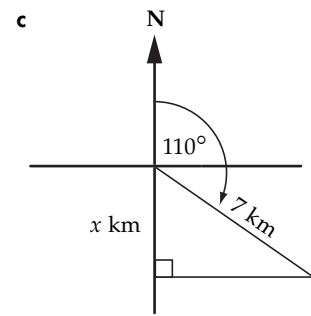
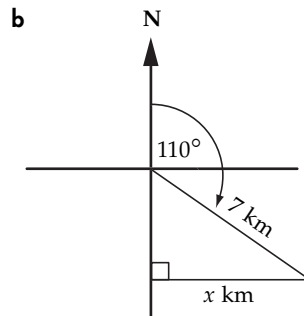
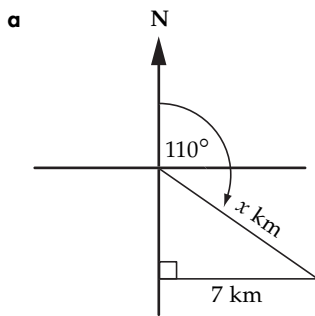
- 2** The bearing of Town L from Town M is 310° . Town L is 63 km from Town M. How far west, to the nearest km, is Town L of Town M? (Let x represent this distance.)



- 3** Penelope is 6 km north and 7 km east of Tom. Determine the bearing of Penelope from Tom to the nearest degree.



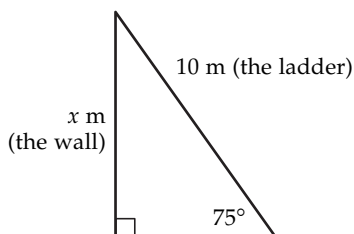
- 4 Uma sailed the first leg of a race on a bearing of 110° for a distance of 7 km. What is the shortest distance (to one decimal place) she has to travel west (x) to be due south of her starting position? (Hint: calculate the size of the angle in the right-angled triangle.)



- 5 To get to an oasis at B from his village at A, Lawrence rode his camel south for 15 km then west for 21 km. Determine the bearing, to the nearest degree, of the oasis from the village at A.
- 6 An orienteer runs due south of the starting point (S) for 8 km. He then runs due west to a tree (T). The bearing of the tree from the starting point is 226° . How far is it from the tree to the starting point? (Answer correct to one decimal place.)
- 7 A plane leaves Darwin and flies on a bearing of 120° . After one hour it is 950 km from Darwin. How far south of Darwin is the plane?
- 8 A new television tower is built 10 km north and 5 km east of a town. Determine the bearing of the tower from the town.
- 9 An orienteer runs on a bearing of 283° . How far has she run when she is 5 km north of the starting point?
- 10 A plane flies from Brisbane on a bearing of 220° at a speed of 250 km per hour. For how long (in hours and minutes) must the plane fly for it to be 750 km south of Brisbane?
- 11 A hill is 7 km north and 3 km west of a town. Determine, to the nearest degree, the bearing of the town from the hill.
- 12 A group of soldiers leaves the base camp and marches 20 km south then 15 km east, where they establish a new camp. Determine the bearing of the base camp from the new camp.
- 13 An island is 15 km north and 12 km east of a port. Determine the bearing of the port from the island.
- 14 A plane leaves Melbourne and flies on a bearing of 280° . It travels at a speed of 950 km per hour. How far west of Melbourne is the plane after 2.5 hours of flying?
- 15 A yacht in distress is 250 nautical miles north and 360 nautical miles west of a rescue ship. Determine the bearing the ship must take to reach the yacht in the shortest time. If the ship maintains a speed of 15 knots (nautical miles per hour) how many hours will it take for the ship to reach the yacht?
- 16 Two planes leave Hobart at the same time. One plane travels on a bearing of 055° and the other plane travels on a bearing of 150° . How far apart are the planes when they are both 250 km due east of Hobart?
- 17 Two ships leave port. The first ship (A) travels on a bearing of 033° and the second ship (B) sails on a bearing of 076° . After 4 hours the ships are both 10 km north of the port. Calculate the distance between them.
- 18 Two ships leave Fremantle, a passenger ship and a freighter. The passenger ship travels on a bearing of 226° . When it has travelled 150 km, the freighter is 200 km north of it. Determine the bearing of the freighter from Fremantle.

14 problems involving trigonometry

e.g. A 10-metre ladder leans against a wall, forming an angle of 75° with the ground. How far up the wall will the ladder reach?



$$\begin{aligned}\sin 75^\circ &= \frac{x}{10} \\ x &= 10 \times \sin 75^\circ \\ \therefore x &= 9.65\dots\end{aligned}$$

$\therefore$ The ladder reaches 9.7 m up the wall.

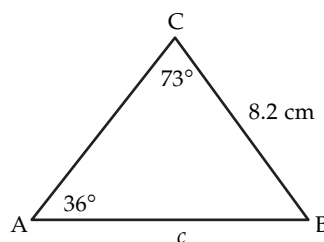
- The diagonal of a rectangle forms an angle of 26° with the longer side. The longer side is 15 cm. Determine the length of the shorter side to the nearest centimetre.
- The shorter side of a rectangle is 15 cm and the diagonal is 24.9 cm. Determine to the nearest degree the angle between the shorter side and the diagonal.
- A 15-metre rope attached to the top of a yacht's mast makes an angle of 77° with the deck. Calculate the height of the mast to one decimal place.
- A wire supporting a radio mast is 27 m long. It makes an angle of 67° with the ground. Determine, to one decimal place, the distance along the ground from the base of the mast to the point the wire is anchored.
- The side of a pyramid makes an angle of 68° with its base, and the perpendicular height of the pyramid is 23.2 m. Determine the slant height to the nearest metre.
- A 15-metre ladder is leaning against a wall. The base of the ladder is 3 m from the wall. Find to the nearest degree the angle the ladder makes with the ground.
- The slant height of a cone is 15 cm and its perpendicular height 11 cm. Calculate to the nearest minute the angle between the side of the cone and its base.
- Lucy wants to clean out the gutter on her house. The gutter is 9.3 m above the ground. Lucy knows that a ladder must extend at least 80 cm beyond a gutter to be safe. If the ladder makes an angle of 70° with the ground, determine the shortest ladder (to the nearest cm) that is safe for Lucy to use.
- The perpendicular height of a cone is 14 cm and the radius of its base is 6 cm. Determine the angle between the side of the cone and its base to the nearest minute.
- An isosceles triangle ABC has sides AB and BC that are both 12 cm in length. The perpendicular from B to the midpoint of AC is 9 cm. Determine the size of the angle ABC to the nearest degree.
- In the rectangle ABCD, $AD = 30$ cm and $\angle ADB = 27^\circ$. Determine the length of AC to the nearest centimetre.
- A rectangle has a diagonal length of 14.5 cm. The diagonal makes an angle of $56^\circ 45'$ with the shorter side. Find the perimeter of the rectangle to the nearest centimetre.
- Drew has a ladder that can extend to 9.5 m. An angle of 12° between the ladder and the wall is considered safe. Can Drew safely use this ladder to access a roof that is 8.5 m above the ground, given that the ladder must also extend 80 cm beyond the roof?
- A surveyor is determining the distance across a canyon. She marks Point A on one side of the canyon and Point B directly opposite it on the other side of the canyon. Then she walks 20 m from Point B to Point C along a line that is perpendicular to the line AB. She measures the angle BCA as $27^\circ 30'$. How wide is the canyon?

You can use the sine rule to solve problems involving triangles that are not right-angled.

In any triangle ABC:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

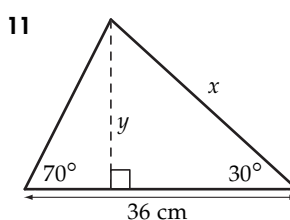
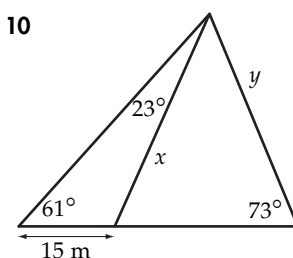
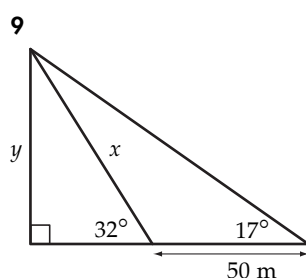
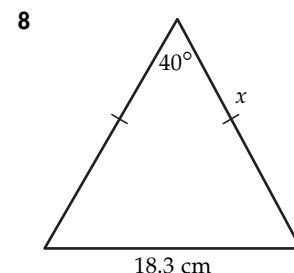
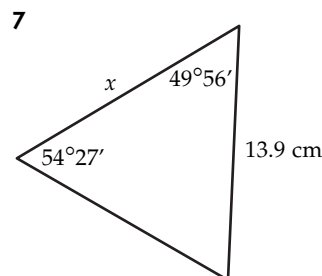
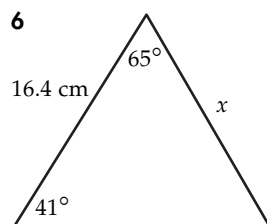
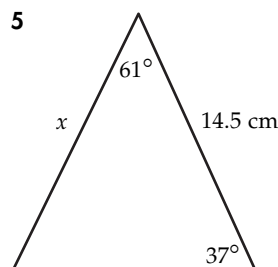
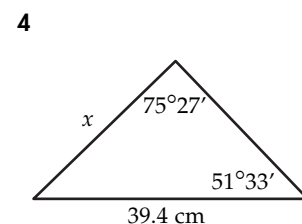
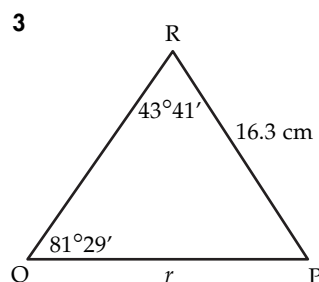
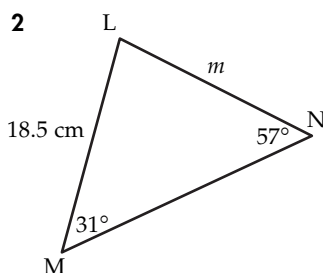
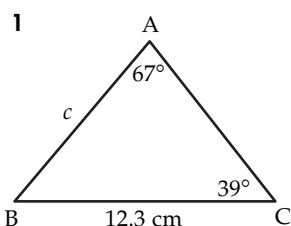
e.g. Find the length of AB.



$$\begin{aligned}\frac{a}{\sin A} &= \frac{c}{\sin C} \\ \frac{8.2}{\sin 36^\circ} &= \frac{c}{\sin 73^\circ} \\ c &= \frac{8.2}{\sin 36^\circ} \times \sin 73^\circ \\ \therefore c &= 13.34\dots\end{aligned}$$

$\therefore$ AB is 13.3 cm (1 d.p.)

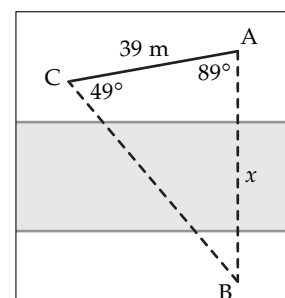
Determine the length of each indicated side, correct to one decimal place.



- 12 A rhombus has sides 15.3 cm long. The acute angle between two adjacent sides is 38° . Determine the length of the shorter diagonal.

- 13 A tightrope walker wants to stretch a rope across a canyon. A and B are the anchor points for the rope. To calculate the length of rope needed, he measures the distance from point A to a tree at point C. Then he measures the angles CAB and ACB. Find the distance from A to B.

- 14 A model plane race starts at Tower 3. Each plane flies to Tower 1, turns through an angle of 56° to reach Tower 2, then turns through an angle of 73° to fly to the finish at Tower 3. The distance from Tower 1 to Tower 2 is 5.3 km. Determine the length of the race to the nearest kilometre.

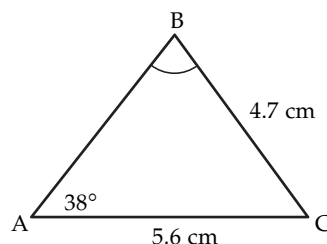


You can use the sine rule to find the size of an angle in a triangle that is not right-angled.

To find an angle, invert the formula:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

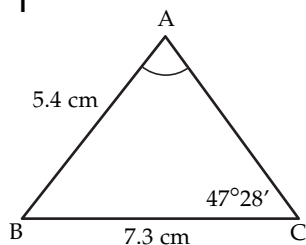
e.g. Find the size of angle B to the nearest minute.



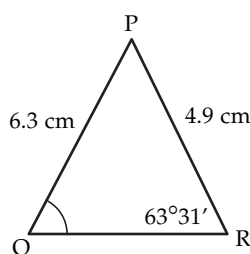
$$\begin{aligned}\frac{\sin B}{5.6} &= \frac{\sin 38^\circ}{4.7} \\ \sin B &= \frac{5.6 \times \sin 38^\circ}{4.7} \\ \sin B &= 0.73355... \\ \therefore B &= 47^\circ 11'\end{aligned}$$

Find the size of each indicated angle to the nearest minute.

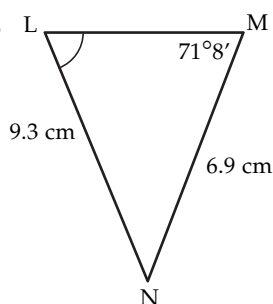
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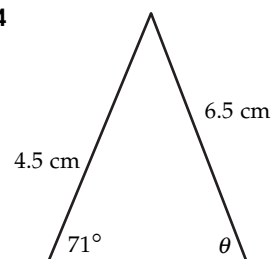
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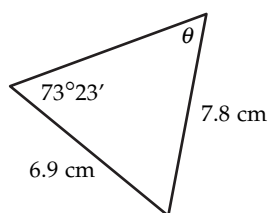
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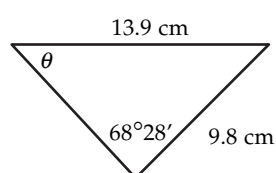
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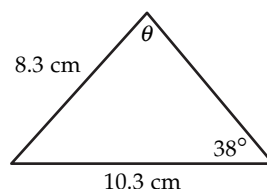
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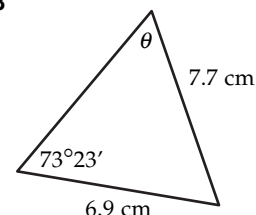
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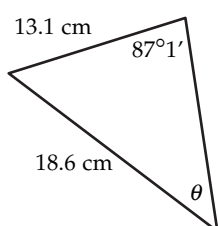
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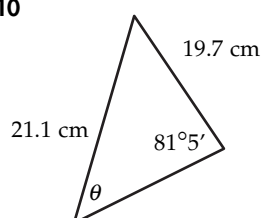
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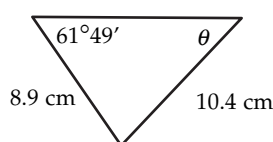
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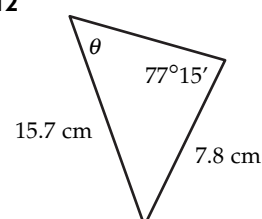
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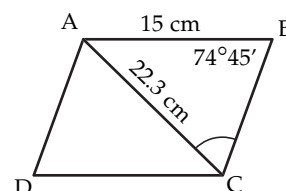


13 In $\triangle LMN$, $\angle L = 38^\circ 49'$, $NM = 4.2$ cm and $LN = 6.3$ cm. Find the size of angle M.

14 In $\triangle XYZ$, $\angle Y = 63^\circ 29'$, $XY = 15.2$ cm and $XZ = 17.3$ cm. Find $\angle XZY$.

15 In $\triangle PQR$, $QR = 18.9$ cm, $PR = 15.3$ cm and $\sin QPR = 0.7$. Determine the size of $\angle RQP$.

16 ABCD is a parallelogram. $AB = 15$ cm, $AC = 22.3$ cm and $\angle ABC = 74^\circ 45'$. Find the size of angle ACB.

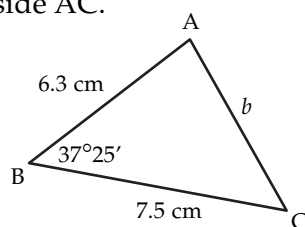


17 Two wires help support a vertical mast. The wires are anchored to the ground on the same side of the mast and in line with the base of the mast. The lengths of the wires are 16.5 m and 13.6 m. The angle between the tops of the two wires is 16° and the wires are 4.3 m apart on the ground. Determine to the nearest degree the angle of elevation of each wire.

In triangle ABC, the cosine rule is: $a^2 = b^2 + c^2 - 2bc \cos A$

This rule can be used to find the length of a side in a triangle that is not right-angled.

e.g. Find the length of the side AC.



$$b^2 = a^2 + c^2 - 2ac \cos B$$

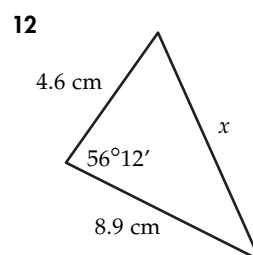
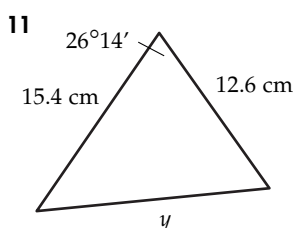
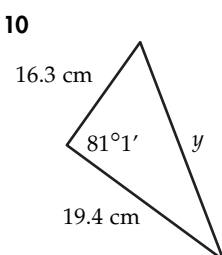
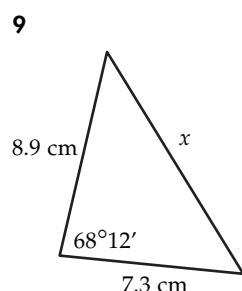
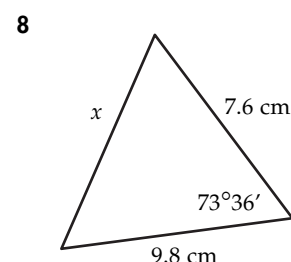
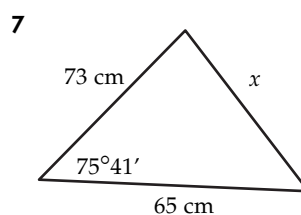
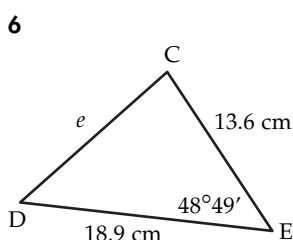
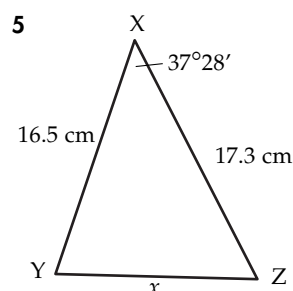
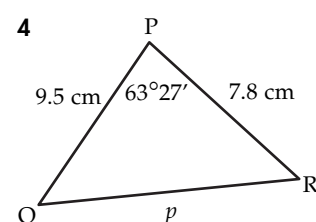
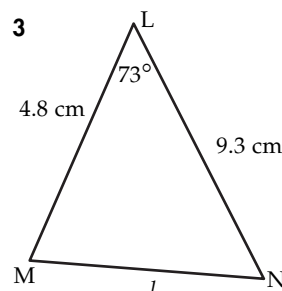
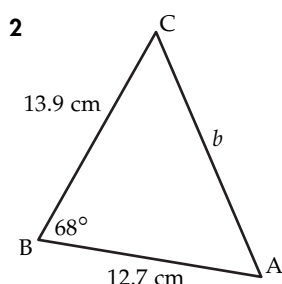
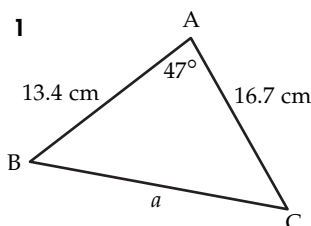
$$b^2 = 7.5^2 + 6.3^2 - 2 \times 7.5 \times 6.3 \times \cos 37^\circ 25'$$

$$b^2 = 20.88 \dots$$

$$\therefore b = 4.56 \dots$$

$$\therefore AC = 4.6 \text{ cm (1 d.p.)}$$

Determine the length of each unknown side correct to one decimal place.



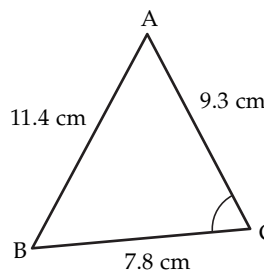
- 13 In $\triangle ABC$, $\angle C = 43^\circ 15'$, $a = 6.9$ cm and $b = 9.7$ cm. Determine the length of side c .
- 14 In $\triangle PQR$, $\angle P = 50^\circ 17'$, $q = 16.2$ cm and $r = 19.4$ cm. Determine the length of side p .
- 15 The acute angle of a parallelogram is 73° and the lengths of the two sides are 16.9 cm and 20.7 cm. Determine the length of the shorter diagonal.
- 16 Quentin and Bill begin walking from the same town. Quentin walks on a bearing of 027° and Bill walks on a bearing of 316° . After 3 hours, Quentin has walked 12.6 km and Bill has walked 11.2 km. What is the distance between them?
- 17 A block of land is triangular in shape, say $\triangle XYZ$, where $\angle X = 68^\circ 15'$, $\angle Y = 33^\circ 19'$, $x = 38.6$ m and $y = 73.6$ m. Fencing costs \$26.30 per metre. How much will it cost to fence the perimeter of the block?

You can use the cosine rule to find the size of an angle in a triangle that is not right-angled. To find an angle, transpose the formula to:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

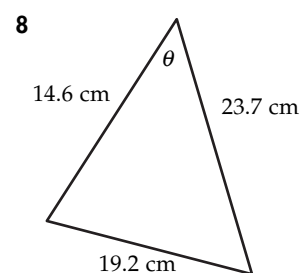
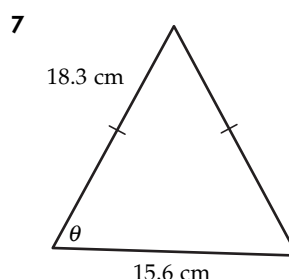
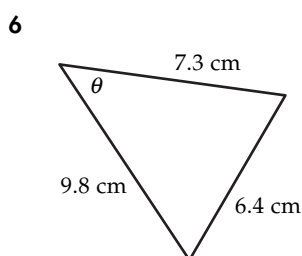
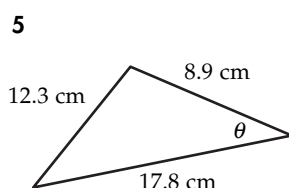
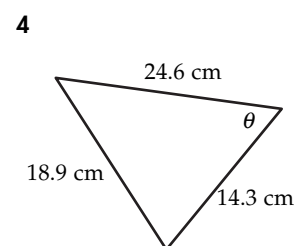
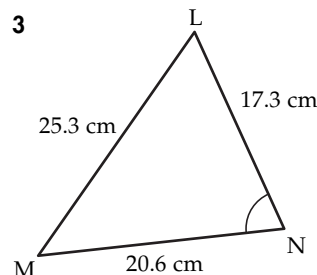
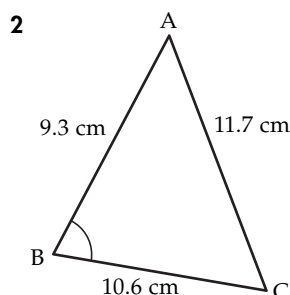
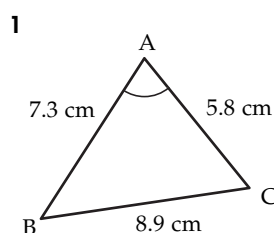
Remember, the smallest side is always opposite the smallest angle; the longest side is always opposite the largest angle.

e.g. In $\triangle ABC$, $a = 7.8$ cm, $b = 9.3$ cm and $c = 11.4$ cm. Find $\angle ACB$ to the nearest minute.



$$\begin{aligned}\cos C &= \frac{a^2 + b^2 - c^2}{2ab} \\ \cos C &= \frac{7.8^2 + 9.3^2 - 11.4^2}{2 \times 7.8 \times 9.3} \\ \cos C &= 0.1197... \\ \therefore C &= 83^\circ 7'\end{aligned}$$

Determine the size of each indicated angle to the nearest minute.



- 9 Sophia said that if you draw a triangle with sides 17.6 cm, 23.4 cm and 18.9 cm, the middle-sized angle is about $52^\circ 37'$. Is she correct?
- 10 In $\triangle LMN$, $l = 63.4$ cm, $m = 59.7$ cm and $n = 43.8$ cm. Calculate the size of the smallest angle.
- 11 Determine the largest angle in a triangle, given that the sides are 7.6 cm, 10.8 cm and 12.7 cm.
- 12 In a rhombus, the length of each side is 10.9 cm and the shorter diagonal is 13.8 cm. Determine the sizes of the angles in the rhombus.
- 13 Two wire supports are attached at the same point on a radio tower. The bases of the two wires and the tower are in a straight line. The wires are 38.5 m and 41.7 m in length and the distance between the two wires along the ground is 4.3 m. Determine the angle of elevation of the longer wire.
- 14 In a parallelogram, the length of the shorter side is 7.6 cm and the diagonals are 11.2 cm and 12.4 cm. Determine the smaller angle between the two diagonals to the nearest minute.
- 15 Two planes leave Sydney at the same time. The first plane flies on a bearing of 073° and the second aeroplane flies in a southerly direction, but not due south. Both planes are flying at 850 km/hr. After $2\frac{1}{2}$ hours they are 2870 km apart. Determine the bearing of the second plane from Sydney to the nearest minute.

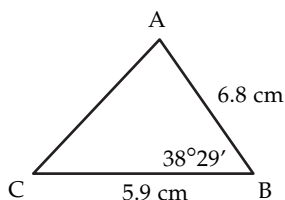
19 area of a triangle

You can use trigonometry to find the area of a triangle, given two sides and the included angle.

In any triangle ABC:

$$\text{Area} = \frac{1}{2}ab \sin C$$

e.g. Determine the area of $\triangle ABC$ to one decimal place.



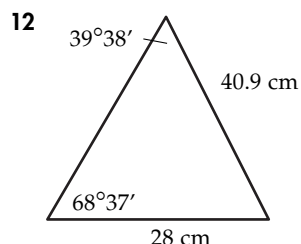
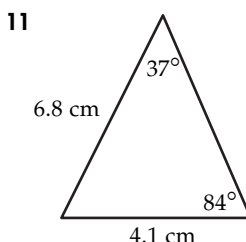
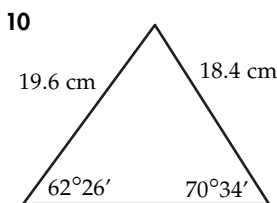
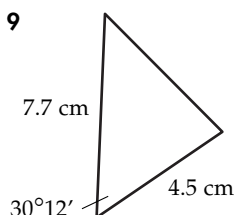
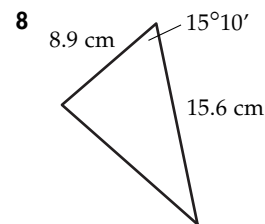
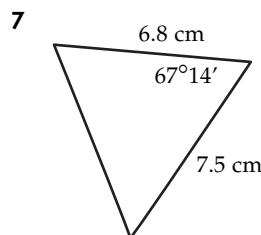
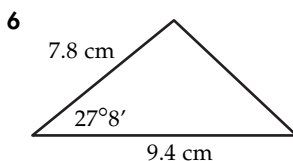
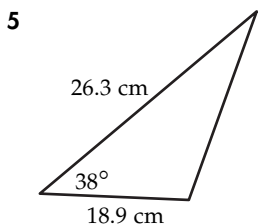
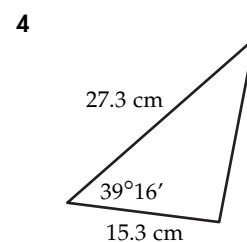
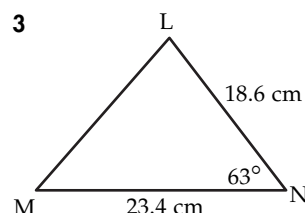
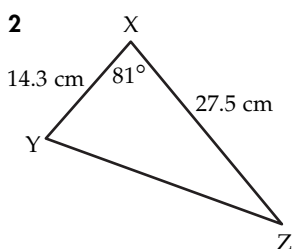
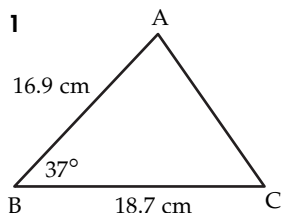
$$\text{Area} = \frac{1}{2}ac \sin B$$

$$\text{Area} = \frac{1}{2} \times 5.9 \times 6.8 \times \sin 38^\circ 29'$$

$$\text{Area} = 12.483\dots$$

$\therefore$ The area of $\triangle ABC$ is 12.5 cm².

Determine the area of each triangle correct to one decimal place.



- 13 In $\triangle LMN$, $\sin N = 0.7321$, $m = 17.3$ cm and $l = 26.3$ cm. Find the area of $\triangle LMN$ to one decimal place.

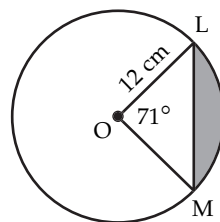
- 14 Determine to one decimal place the area of triangle PQR given that $\angle Q = 63^\circ 28'$, $r = 13.74$ cm and $p = 15.8$ cm.

- 15 Find the area of triangle PQR to one decimal place, given that the sides are 17.3 cm, 18.6 cm and 19.4 cm and the smallest angle is $54^\circ 7'$.

- 16 The area of $\triangle ABC$ is 22.4 cm². Determine the length of side AB to one decimal place, given that $a = 10.3$ cm and $\angle B = 29^\circ 14'$.

- 17 The area of a triangle is 211.2 cm². The lengths of two of the sides are 27.3 cm and 18.6 cm. Determine to the nearest minute the size of the acute angle between these two sides.

- 18 You can use trigonometry to find the area of a segment of a circle. To find the area of the minor segment (shaded):



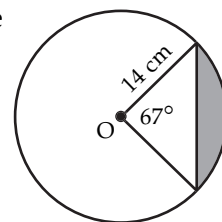
- a find the area of sector LMO, to 2 decimal places.

$$\left(\text{Area} = \frac{\theta}{360} \times \pi r^2 \right)$$

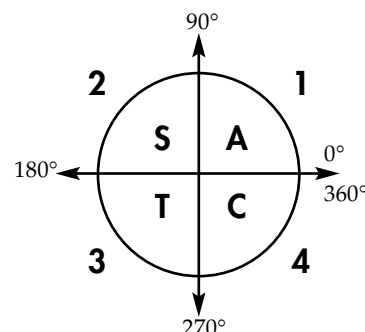
- b find the area of $\triangle LMO$, to 2 decimal places.

- c area of segment = area of sector – area of triangle
Answer to 1 decimal place.

- 19 Determine the area of the shaded segment to one decimal place.



The trigonometric ratios of any angle greater than 90° can be expressed as the ratio of an acute angle. You need to determine the acute angle between the given angle and the horizontal; that is, 180° or 360° . The diagram at right is useful for determining the ratios of any angle because it indicates which ratio is positive. All ratios are positive in quadrant 1; only sine is positive in quadrant 2, cos and tan are negative; only tan is positive in quadrant 3, sin and cos are negative; and only cos is positive in quadrant 4. Note that the reciprocal ratios take the same sign as the basic ratios.



e.g. 1 $\sin 170^\circ = \sin(180 - 170)^\circ$
 $= \sin 10^\circ$

2 $\tan 300^\circ = \tan(360 - 300)^\circ$
 $= -\tan 60^\circ$

3 $\sec 230^\circ = \sec(180 + 50)^\circ$
 $= -\sec 50^\circ$

4 $\tan 400^\circ = \tan(360 + 40)^\circ$
 $= \tan 40^\circ$

Determine the sign (positive or negative) of each ratio.

- | | | | |
|-----------------------------------|--------------------|--------------------|--------------------|
| 1 $\sin 127^\circ$ | 2 $\cos 330^\circ$ | 3 $\tan 150^\circ$ | 4 $\cos 125^\circ$ |
| 5 $\operatorname{cosec} 67^\circ$ | 6 $\cot 359^\circ$ | 7 $\sec 250^\circ$ | 8 $\sin 277^\circ$ |

Use a calculator to determine each ratio to 4 decimal places. (The calculator will give you the sign.)

- | | | | |
|---------------------|---------------------|---------------------|---------------------|
| 9 $\sin 230^\circ$ | 10 $\cos 123^\circ$ | 11 $\tan 317^\circ$ | 12 $\cos 280^\circ$ |
| 13 $\tan 115^\circ$ | 14 $\sin 123^\circ$ | 15 $\cos 200^\circ$ | 16 $\tan 200^\circ$ |
| 17 $\tan 380^\circ$ | 18 $\cos 700^\circ$ | 19 $\tan 520^\circ$ | 20 $\sin 370^\circ$ |

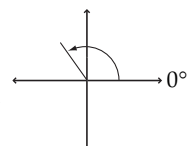
Complete the following.

- | | |
|--|--|
| 21 $\sin 130^\circ = \sin(180 - \theta)^\circ$
$\therefore \theta =$ | 22 $\cos 220^\circ = -\cos(180 + \theta)^\circ$
$\therefore \theta =$ |
| 23 $\tan 325^\circ = -\tan(360 - \theta)^\circ$
$\therefore \theta =$ | 24 $\operatorname{cosec} 260^\circ = -\operatorname{cosec}(180 + \theta)^\circ$
$\therefore \theta =$ |

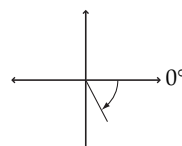
Give each ratio as a ratio of the appropriate acute angle. (Don't forget to include a + or - sign.)

- | | | | |
|---------------------|---------------------|---------------------|-------------------------------------|
| 25 $\cos 326^\circ$ | 26 $\sin 140^\circ$ | 27 $\tan 220^\circ$ | 28 $\tan 340^\circ$ |
| 29 $\cos 130^\circ$ | 30 $\sec 150^\circ$ | 31 $\cot 327^\circ$ | 32 $\operatorname{cosec} 130^\circ$ |
| 33 $\sin 400^\circ$ | 34 $\sin 440^\circ$ | 35 $\cos 480^\circ$ | 36 $\tan 760^\circ$ |

To form a **positive** angle, start at 0° and rotate **anticlockwise**.



To form a **negative** angle, start at 0° and rotate **clockwise**.



Note:

$$\begin{aligned}\sin(-70^\circ) &= \sin 290^\circ \\ &= \sin(360 - 70)^\circ \\ &= -\sin 70^\circ\end{aligned}$$

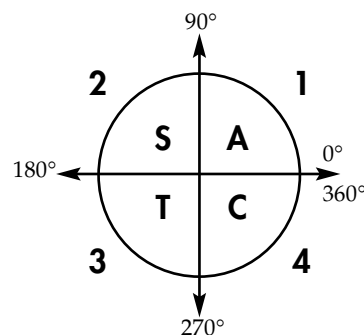
Give the equivalent positive angle for each of these negative angles.

- | | | | |
|----------------|-----------------|----------------|-----------------|
| 37 -80° | 38 -100° | 39 -60° | 40 -170° |
|----------------|-----------------|----------------|-----------------|

Write as the ratio of a positive angle.

- | | | | |
|----------------------|----------------------|-----------------------|-----------------------|
| 41 $\cos(-30^\circ)$ | 42 $\tan(-50^\circ)$ | 43 $\sin(-150^\circ)$ | 44 $\sec(-140^\circ)$ |
|----------------------|----------------------|-----------------------|-----------------------|

The trigonometric ratios of any angle can be expressed as the ratio of the appropriate acute angle. Trigonometric ratios are either positive or negative. This diagram shows the positive ratios for each quadrant. In quadrant 1 all ratios are positive; in quadrant 2 only sine is positive, cos and tan are negative; in quadrant 3 only tan is positive; and in quadrant 4 only cos is positive. The reciprocal ratios take the same sign as the basic ratios.



e.g. Determine the value of θ if $0^\circ \leq \theta \leq 360^\circ$.

1 $\cos \theta = 0.3$

cos is positive in quadrants 1 and 4

$$\theta = 72^\circ 33' \text{ or } 360^\circ - 72^\circ 33'$$

$$\therefore \theta = 72^\circ 33' \text{ or } 287^\circ 27'$$

2 $\tan \theta = -2.3$

tan is negative in quadrants 2 and 4

$$\theta = 180^\circ - 66^\circ 30' \text{ or } 360^\circ - 66^\circ 30'$$

$$\therefore \theta = 113^\circ 30' \text{ or } 293^\circ 30'$$

3 $\sec \theta = 2.0$

as cos is positive in quadrants 1 and 4, so is sec

$$\cos \theta = \frac{1}{2.0}$$

$$\theta = 60^\circ \text{ or } 360^\circ - 60^\circ$$

$$\therefore \theta = 60^\circ \text{ or } 300^\circ$$

Give the quadrants in which θ will occur.

1 $\sin \theta = -0.3$

2 $\cos \theta = -0.6$

3 $\sin \theta = 0.6$

4 $\cos \theta = -0.4$

5 $\tan \theta = 1.6$

6 $\tan \theta = 0.7$

7 $\tan \theta = -4.3$

8 $\operatorname{cosec} \theta = 1.2$

9 $\sec \theta = 2.3$

10 $\cot \theta = 1.6$

11 $\operatorname{cosec} \theta = -1.3$

12 $\sec \theta = -1.5$

Given that θ is acute, determine the value of each angle to the nearest degree.

13 $\sin \theta = 0.415$

14 $\cos \theta = 0.563$

15 $\tan \theta = 1.5$

16 $\sec \theta = 1.3$

17 $\operatorname{cosec} \theta = 2.513$

18 $\cot \theta = 0.85$

19 $\sin \theta = 0.691$

20 $\cot \theta = 0.45$

Given that θ is between 0° and 360° , determine the value of θ to the nearest degree.

21 $\cos \theta = 0.6$

22 $\sin \theta = -0.35$

23 $\tan \theta = 1.55$

24 $\tan \theta = -3.41$

25 $\cos \theta = -0.351$

26 $\sin \theta = 0.784$

27 $\cos \theta = 0.37$

28 $\sin \theta = -0.1327$

29 $\tan \theta = 0.9$

30 $\cos \theta = -0.72$

31 $\tan \theta = -0.63$

32 $\sin \theta = 0.583$

33 $\sec \theta = -1.3$

34 $\operatorname{cosec} \theta = 1.6$

35 $\cot \theta = 0.8$

36 $\operatorname{cosec} \theta = -4.3$

37 $\sec \theta = 2.3$

38 $\cot \theta = -1.4$

39 $\sec \theta = -3.1$

40 $\operatorname{cosec} \theta = 3.4$

Given that $0^\circ \leq \theta \leq 360^\circ$, find the value of θ to the nearest degree.

41 $\sin \theta = 0.41$ and $\cos \theta$ is positive

42 $\tan \theta = -1.67$ and $\sin \theta$ is negative

43 $\cos \theta = 0.83$ and $\operatorname{cosec} \theta < 0$

44 $\cos \theta = 0.6$ and $\tan \theta < 0$

45 $\tan \theta = -3.67$ and $\operatorname{cosec} \theta > 0$

46 $\sin \theta = -0.308$ and $\cot \theta > 0$

The sine rule, cosine rule and area of a triangle formula are usually applied to triangles that are not right-angled. For any triangle ABC:

The sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\text{or } \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

The cosine rule

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\text{or } a^2 = b^2 + c^2 - 2bc \cos A$$

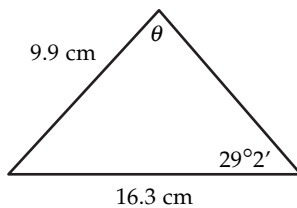
Area of a triangle

$$A = \frac{1}{2}ab \sin C$$

When using the **sine** rule, take care when finding the size of an obtuse angle as most calculators only give acute angle answers. It helps to draw a sketch of the triangle to see if there is an obtuse angle (one side much longer than the other sides is a good indication).

When using the **cosine** rule, if an angle is obtuse the calculator will show a negative cosine ratio.

e.g. 1 Find the size of θ .



Note: From the sketch θ is obtuse

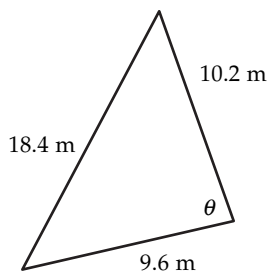
$$\frac{\sin \theta}{16.3} = \frac{\sin 29^\circ 2'}{9.9}$$

$$\sin \theta = \frac{16.3 \times \sin 29^\circ 2'}{9.9}$$

$$\therefore \theta = 53^\circ 2'$$

The required angle is obtuse, so the correct value of θ is: $180^\circ - 53^\circ 2' = 126^\circ 58'$

2 Find θ to the nearest minute.



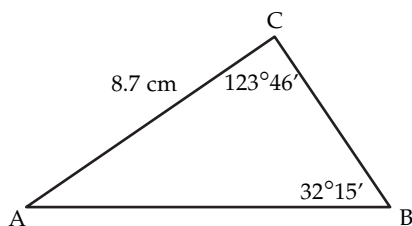
$$\cos \theta = \frac{10.2^2 + 9.6^2 - 18.4^2}{2 \times 10.2 \times 9.6}$$

$$\cos \theta = -0.7268\dots$$

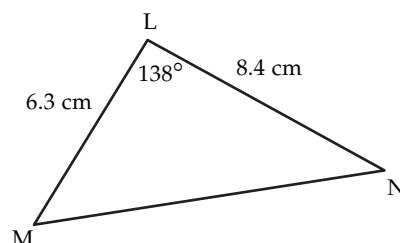
$$\therefore \theta = 136^\circ 38'$$

In the following questions, determine angles to the nearest minute, lengths to one decimal place and areas to one decimal place.

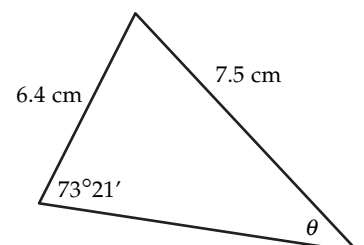
1 Determine the length AB.



2 Find the area.

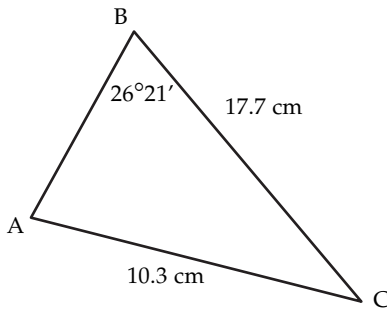


3 Determine the value of θ .

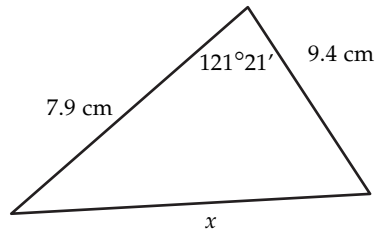


In the following questions, determine angles to the nearest minute, lengths to one decimal place and areas to one decimal place.

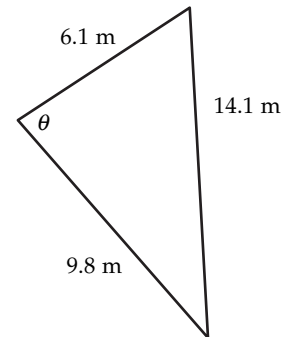
- 4 Find the size of angle A.



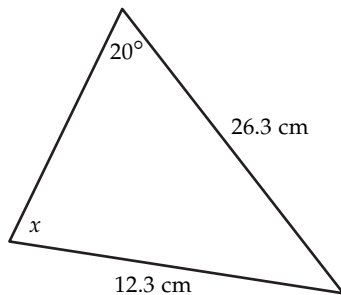
- 5 Determine the length of x .



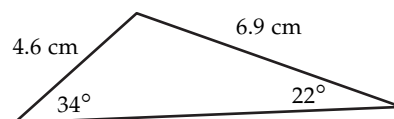
- 6 Find the value of θ .



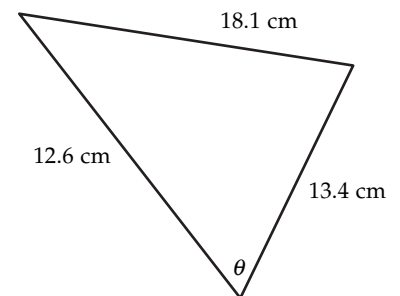
- 7 Find the value of x .



- 8 Find the area.



- 9 Find the value of θ .

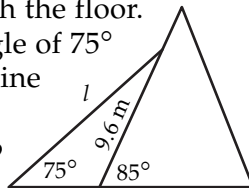


- 10 In $\triangle PQR$, $p = 17.3$ cm, $q = 10.7$ cm and $r = 8.9$ cm. Determine the size of $\angle QPR$.

- 11 In $\triangle LMN$, $l = 29.4$ cm, $n = 16.3$ cm and $\angle N = 29^\circ$. Determine the value of $\angle LMN$.

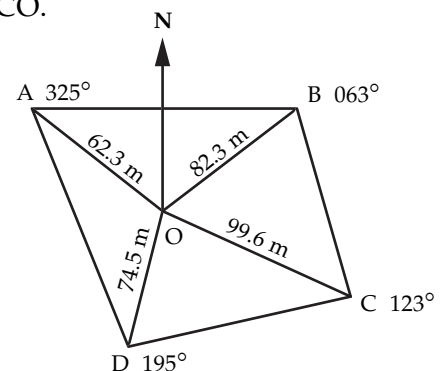
- 12 In $\triangle XYZ$, $x = 27.3$ cm, $z = 16.8$ cm and $\angle XYZ = 23^\circ 4'$. Determine the size of the largest angle.

- 13 A builder leans a ladder against the side of an A-framed house. The sides of the house make an angle of 85° with the floor. The ladder makes an angle of 75° with the ground. Determine the length of the ladder needed to reach 9.6 m up the side of the house.

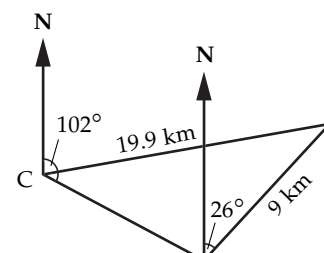


- 14 Two trawlers are in a direct line from a lighthouse. From the top of the lighthouse, the angles of depression of the two trawlers are $25^\circ 8'$ and $5^\circ 23'$ respectively. The trawlers are 350 m apart. Determine the distance from the lighthouse to the nearest trawler and the height of the lighthouse.

- 15 This diagram represents a plane table survey. It shows the bearings of points A, B, C and D. Determine the area bounded by ABCO.

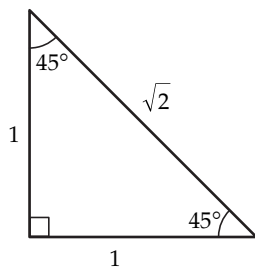
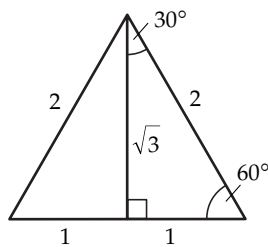


- 16 A ship leaves Cairns sailing on a bearing of 102° . After some time it changes course and sails on a bearing of 026° . It travels 9 km and stops when it is 19.9 km from Cairns. Find the new bearing of the ship from Cairns.



23 exact ratios of 30°, 45° and 60°

The exact ratios of 30°, 60° and 45° can be determined by using these triangles.



These reciprocal ratios may be useful:

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

Use the triangles above to complete these tables.

1		30°	60°	45°
sin				
cos				
tan				

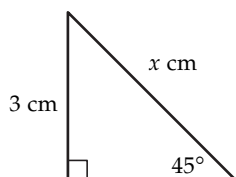
2		30°	60°	45°
cosec				
sec				
cot				

Determine the exact ratio, giving the answer in simplest form.

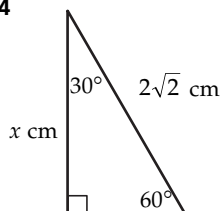
- | | | | |
|--|--|---|--|
| 3 $\sec 45^\circ$ | 4 $\sec 30^\circ$ | 5 $\operatorname{cosec} 60^\circ$ | 6 $\sin 30^\circ + \cos 60^\circ$ |
| 7 $\sin 30^\circ \cdot \cos 30^\circ$ | 8 $\sin^2 45^\circ$ | 9 $\sin 30^\circ + \sin 60^\circ$ | 10 $\tan 30^\circ \cdot \tan 60^\circ$ |
| 11 $\sin 30^\circ + \cos^2 45^\circ$ | 12 $\sin^2 30^\circ + \cos^2 30^\circ$ | 13 $\cos 30^\circ \cdot \sin 60^\circ$ | 14 $\frac{\sin 60^\circ}{\cos 30^\circ}$ |
| 15 $\frac{\cos 60^\circ}{\sin 60^\circ}$ | 16 $2\cos^2 30^\circ$ | 17 $\operatorname{cosec}^2 45^\circ$ | 18 $\cot^2 60^\circ$ |
| 19 $\sec^2 45^\circ$ | 20 $3\tan 30^\circ$ | 21 $\sec^2 60^\circ$ | 22 $\tan^2 45^\circ \cdot \operatorname{cosec} 60^\circ$ |
| 23 $\frac{\sin 30^\circ}{\tan 30^\circ}$ | 24 $\cos^2 30^\circ + \cos^2 60^\circ$ | 25 $\operatorname{cosec}^2 45^\circ - 1$ | 26 $\frac{\cos 30^\circ + \sin 60^\circ}{\cos 60^\circ}$ |
| 27 $\operatorname{cosec}^2 45^\circ + \sin 30^\circ \cdot \operatorname{cosec} 30^\circ$ | 28 $\frac{\sin 30^\circ \cdot \cos 60^\circ}{\tan 30^\circ}$ | 29 $\frac{\cos^2 45^\circ}{\sin 30^\circ + \cos 60^\circ}$ | |
| 30 $\cot^2 30^\circ - 2\tan^2 45^\circ$ | 31 $\sin^2 30^\circ + 2\cos 60^\circ$ | 32 $\frac{1}{2}\sec^2 45^\circ - \frac{\sin^2 60^\circ}{3}$ | |

Determine the exact value of x .

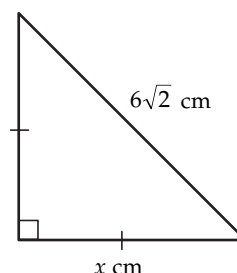
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34



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- 36 Show that $\sin 30^\circ \cdot \cos^2 45^\circ = \frac{1}{4}(\sin^2 60^\circ + \sin^2 45^\circ \cos^2 45^\circ)$
- 37 Show that $\cot 60^\circ \cdot \operatorname{cosec} 60^\circ = \cos^2 60^\circ + \sin^2 60^\circ - \tan^2 30^\circ$
- 38 Show that $(\tan 30^\circ \cdot \sec 45^\circ)^2 = \sin 60^\circ \cdot \tan 30^\circ + \frac{1}{2}\cot^2 60^\circ$

Reciprocal ratios

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad \sec \theta = \frac{1}{\cos \theta} \quad \cot \theta = \frac{1}{\tan \theta}$$

Complementary ratios

$$\begin{aligned} \sin \theta &= \cos(90 - \theta) \quad \text{and} \quad \cos \theta = \sin(90 - \theta) \\ \operatorname{cosec} \theta &= \sec(90 - \theta) \quad \text{and} \quad \sec \theta = \operatorname{cosec}(90 - \theta) \\ \tan \theta &= \cot(90 - \theta) \quad \text{and} \quad \cot \theta = \tan(90 - \theta) \end{aligned}$$

The Pythagorean results

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta \quad \frac{\cos \theta}{\sin \theta} = \cot \theta$$

e.g. Simplify each expression.

1 $\cot A \sin A$

$$\begin{aligned} \cot A \sin A &= \frac{\cos A}{\sin A} \cdot \frac{\sin A}{1} \\ &= \cos A \end{aligned}$$

2 $\frac{\sin \theta \sec \theta}{\tan \theta}$

$$\begin{aligned} \frac{\sin \theta \sec \theta}{\tan \theta} &= \sin \theta \times \sec \theta \div \tan \theta \\ &= \sin \theta \times \frac{1}{\cos \theta} \div \frac{\sin \theta}{\cos \theta} \\ &= \frac{\sin \theta}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} \\ &= 1 \end{aligned}$$

3 $\sec^2 \theta - 1$

$$\begin{aligned} \sec^2 \theta - 1 &= 1 + \tan^2 \theta - 1 \\ &= \tan^2 \theta \end{aligned}$$

Simplify each expression.

1 $1 - \sin^2 \theta$

2 $\operatorname{cosec}^2 \theta - 1$

3 $\operatorname{cosec}^2 \theta - \cot^2 \theta$

4 $\sec^2 \theta - \tan^2 \theta$

5 $\sin A \cdot \sec A$

6 $\sin \theta \operatorname{cosec} \theta$

7 $\tan \theta \cos \theta$

8 $\sin \alpha \cos \alpha \tan \alpha$

9 $\operatorname{cosec} \theta \cot \theta \cos \theta$

10 $\cos A \sec A \operatorname{cosec} A$

11 $\cos(90 - \beta) \cdot \operatorname{cosec} \beta$

12 $\frac{\sin \theta}{\sin(90 - \theta)}$

13 $\tan \theta \cdot \tan(90 - \theta)$

14 $\cot \theta \operatorname{cosec} \theta \tan \theta$

15 $\operatorname{cosec}(90 - \alpha) \cdot \cos \alpha$

16 $\frac{\cos A}{\sec A}$

17 $\frac{\sin A}{\operatorname{cosec} A}$

18 $3 \sin^2 \theta + 3 \cos^2 \theta$

19 $5 - 5 \cos^2 \theta$

20 $4 - 4 \sin^2 \beta$

21 $(1 - \sin \beta)(1 + \sin \beta)$

22 $\cot A \cdot \cos A \cdot \tan A$

23 $\cos \theta \tan \theta \operatorname{cosec} \theta$

24 $(\sin \theta + \cos \theta)^2 - 1$

25 $\sin \alpha \cdot \sqrt{\operatorname{cosec}^2 \alpha - 1}$

26 $2 + \cot^2 \theta + \tan^2 \theta$

27 $\frac{1}{\sec \theta \cdot \cos \theta} + \frac{1}{\operatorname{cosec} \theta \cdot \sin \theta}$

28 $\sin \theta \cot \theta \cos \theta + \frac{1}{1 + \cot^2 \theta}$

29 $\frac{1}{\sec^2 \theta} + \frac{1}{\operatorname{cosec}^2 \theta} + \cot^2 \theta$

30 $\cot \theta \tan \theta - \frac{1}{\sec \theta} \cdot \cos \theta$

Reciprocal ratios

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

The Pythagorean results

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$\frac{\cos \theta}{\sin \theta} = \cot \theta$$

e.g. 1 Prove $\sin \theta \cdot \cot \theta \cdot \sec \theta = 1$

$$\begin{aligned} \text{L.H.S.} &= \sin \theta \cdot \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\cos \theta} \\ &= 1 \\ &= \text{R.H.S.} \end{aligned}$$

2 Prove $\sin^2 \theta + \cos^2 \theta = \tan \theta \cot \theta$

$$\begin{aligned} \text{L.H.S.} &= 1 & \text{R.H.S.} &= \frac{\sin \theta}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} \\ & & &= 1 \end{aligned}$$

$$\therefore \text{L.H.S.} = \text{R.H.S.}$$

Prove each identity.

1 $\cot \theta \sin \theta = \cos \theta$

3 $\sec \theta \cot \theta = \operatorname{cosec} \theta$

5 $\sec \theta \tan \theta \operatorname{cosec} \theta = \sec^2 \theta$

7 $\operatorname{cosec} \theta \sqrt{1 - \sin^2 \theta} = \cot \theta$

9 $\sin^2 \beta = 1 - \frac{1}{\sec^2 \beta}$

11 $\cot^2 \theta \cos^2 \theta = \cot^2 \theta - \cos^2 \theta$

13 $\sqrt{1 + \cot^2 X} = \cot X \sec X$

15 $\cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1$

17 $\cos \theta \sin \theta \cot \theta = 1 - \sin^2 \theta$

19 $(1 - \sin^2 \alpha) \operatorname{cosec}^2 \alpha = \cot^2 \alpha$

21 $(\sin \beta + \cos \beta)^2 = 1 + 2 \sin \beta \cos \beta$

23 $\frac{\cot^2 \theta}{1 + \cot^2 \theta} = 1 - \sin^2 \theta$

2 $\cos \theta \sec \theta = \cot \theta$

4 $\tan \theta \operatorname{cosec} \theta = \sec \theta$

6 $\frac{\cot \theta}{\tan \theta} = \cot^2 \theta$

8 $\sqrt{1 - \cos^2 \alpha} = \sin \alpha$

10 $\sin A \cos A \cot A = \cos^2 A$

12 $(\sin \theta + \cos \theta) \operatorname{cosec} \theta \cdot \sec \theta = \frac{1}{\sin \theta} + \frac{1}{\cos \theta}$

14 $\cos^2 \alpha - \sin^2 \alpha = 1 - 2 \sin^2 \alpha$

16 $\sin^2 A (1 + \cot^2 A) = 1$

18 $(1 - \cos^2 \beta) \sec^2 \beta = \tan^2 \beta$

20 $\frac{1 + \cot^2 \theta}{\sec^2 \theta - \tan^2 \theta} = \frac{1}{\sin^2 \theta}$

22 $\frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta} = 2 \operatorname{cosec}^2 \theta$

24 $\tan^2 \theta \cdot \sin^2 \theta = \tan^2 \theta - \sin^2 \theta$

26 trigonometric equations II

e.g. 1 Solve to the nearest minute.

$$\tan \theta = -0.3564 \text{ for } 0^\circ \leq \theta \leq 360^\circ$$

The negative sign shows that the required angles are in quadrants 2 and 4.

$$\therefore \theta = (180^\circ - 19^\circ 37') \text{ or } (360^\circ - 19^\circ 37')$$

$$\therefore \theta = 160^\circ 23' \text{ or } 340^\circ 23'$$

2 Solve $2\sin^2 x - 5\sin x + 2 = 0$ given that $0^\circ \leq \theta \leq 360^\circ$

$$2\sin^2 x - 5\sin x + 2 = 0$$

$$(2\sin x - 1)(\sin x - 2) = 0$$

$$\therefore \sin x = \frac{1}{2} \text{ or } \sin x = 2$$

but as $-1 \leq \sin x \leq 1$ $\sin x = 2$ is impossible, the only solution is $\sin x = \frac{1}{2}$

$$\therefore x = 30^\circ \text{ or } (180 - 30)^\circ$$

$$\therefore x = 30^\circ \text{ or } 150^\circ$$

Note: Solutions to this type of equation often involve the exact ratios of 30° , 45° and 60° .

Solve each equation to the nearest minute (where necessary) given that $0^\circ \leq \theta \leq 360^\circ$.

1 $\cos \theta = 0.732$

2 $\sin \theta = 0.7351$

3 $\tan \theta = 0.356$

4 $\sin \theta = -0.563$

5 $\cos \theta = -0.374$

6 $\tan \theta = -1.236$

7 $\cos \theta = -0.861$

8 $\tan \theta = -6.3$

9 $\sec \theta = 1.43$

10 $\cot \theta = 2.45$

11 $\operatorname{cosec} \theta = -1.577$

12 $\cos \theta = \frac{1}{2}$

13 $\tan \theta = \frac{1}{\sqrt{3}}$

14 $\sin \theta = -\frac{1}{\sqrt{2}}$

15 $\cos \theta = \frac{\sqrt{3}}{2}$

16 $2\cos \theta = \frac{1}{2}$

17 $\sin^2 \theta = \frac{1}{2}$

18 $3\sin \theta = 1$

19 $2\sin \theta + 1 = \frac{1}{2}$

20 $\operatorname{cosec} \theta = \frac{2}{\sqrt{3}}$

21 $\sec \theta = -2$

22 $\cot \theta = -\sqrt{3}$

23 $2\tan \theta = 0.634$

24 $3\sin \theta - 2 = 0$

25 $\tan^2 \theta + 3\tan \theta + 2 = 0$

26 $\tan^2 \theta + \tan \theta - 2 = 0$

27 $6\sin^2 \theta + \sin \theta - 1 = 0$

28 $2\cos^2 \theta + 3\cos \theta - 2 = 0$

29 $3\sin^2 \theta - \sin \theta - 2 = 0$

30 $\sin 2\theta = \frac{1}{2}$

31 $\tan 2\theta = \sqrt{3}$

32 $\cos 2\theta = -0.6$

33 $\sin 2\theta = -0.731$

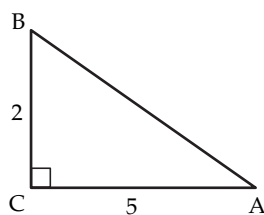
34 $\sin 3\theta = \frac{\sqrt{3}}{2}$

35 $\cos \frac{1}{2}\theta = \frac{1}{\sqrt{2}}$

36 $\sin \frac{1}{2}\theta = \frac{1}{2}$

e.g.

- 1** Given that $\tan A = \frac{2}{5}$ and $\angle A$ is acute, find the exact value of $\cos A$ and $\operatorname{cosec} A$.



$$AB^2 = AC^2 + BC^2$$

$$AB^2 = 5^2 + 2^2$$

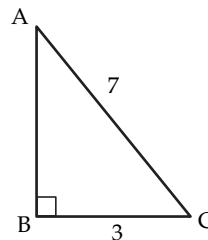
$$AB^2 = 29$$

$$\therefore AB = \sqrt{29}$$

$$\therefore \cos A = \frac{5}{\sqrt{29}} \text{ and}$$

$$\operatorname{cosec} A = \frac{\sqrt{29}}{2}$$

- 2** Given that $\sin A = \frac{3}{7}$ and $\cos A < 0$ find the exact value of $\tan A$ and $\sec A$.



$$AC^2 = AB^2 + BC^2$$

$$49 = AB^2 + 9$$

$$AB^2 = 40$$

$$\therefore AB = \sqrt{40}$$

Since $\sin A > 0$ and $\cos A < 0$, angle A must be in quadrant 2. The correct sign must be applied. $\therefore \tan A = -\frac{3}{\sqrt{40}}$ and $\sec A = -\frac{7}{\sqrt{40}}$

It is also possible to use the identity $\sin^2 A + \cos^2 A = 1$ to find these ratios.

Determine the exact value for each ratio given that the angle is acute.

- | | |
|--|--|
| 1 $\tan A = \frac{3}{4}$, find $\sin A$ and $\cos A$ | 2 $\sin \theta = \frac{3}{4}$, find $\cos \theta$ and $\tan \theta$ |
| 3 $\cos X = \frac{1}{6}$, find $\sin X$ and $\tan X$ | 4 $\tan A = \frac{4}{7}$, find $\cos A$ and $\operatorname{cosec} A$ |
| 5 $\sin \beta = \frac{2}{7}$, find $\cos \beta$ and $\tan \beta$ | 6 $\tan \alpha = \frac{6}{7}$, find $\sin \alpha$ and $\sec \alpha$ |
| 7 $\cos B = \frac{3}{\sqrt{11}}$, find $\sin B$ and $\cot B$ | 8 $\sin A = \frac{2}{\sqrt{13}}$, find $\cot A$ and $\sec A$ |
| 9 $\tan A = \frac{3}{\sqrt{2}}$, find $\sin A$ and $\sec A$ | 10 $\sin C = \frac{\sqrt{3}}{5}$, find $\cos C$ and $\cot C$ |
| 11 $\cos \theta = t$, find $\sin \theta$ and $\tan \theta$ in terms of t | 12 $\sin \theta = \frac{1}{t}$, find $\cos \theta$ and $\cot \theta$ in terms of t |
| 13 If $\cos \chi = \frac{4}{11}$, show that $\sin^2 \chi + \cos^2 \chi = 1$ | |
| 14 If $\tan \theta = \frac{3}{5}$, show that $1 + \cos^2 \theta = \operatorname{cosec}^2 \theta$ | |
| 15 If $\sin X = \frac{7}{8}$, show that $1 + \tan^2 X = \sec^2 X$ | |

Determine the exact value for each ratio. Each angle lies between 0° and 360° .

- 16** $\cos \theta = -\frac{\sqrt{3}}{4}$ and $\sin \theta > 0$, find $\operatorname{cosec} \theta$ and $\cot \theta$
- 17** $\sin A = \frac{3}{7}$ and $\cos A > 0$, find $\sec A$ and $\tan A$
- 18** $\cos \alpha = -\frac{3}{7}$ and $\tan \alpha < 0$, find $\cot \alpha$ and $\sin \alpha$
- 19** $\cot B = \frac{5}{6}$ and $\sin B < 0$, find $\sin B$ and $\sec B$
- 20** $\operatorname{cosec} \theta = -\frac{4}{3}$ and $\cos \theta < 0$, find $\cot \theta$ and $\sec \theta$
- 21** $\tan \theta = \frac{5}{\sqrt{7}}$ and $\cos \theta > 0$, find $\sin \theta$ and $\sec \theta$

Use your calculator to find the ratios of these boundary angles. A display of E on your calculator means that the ratio is undefined; the reciprocal of an undefined ratio is 0.

e.g. Find the value of $\operatorname{cosec} 90^\circ$.

The value of $1 \div \sin 90^\circ$ from the calculator is 1.

Remember, two angles are complementary if they add to 90° .

Use your calculator to complete these tables.

1

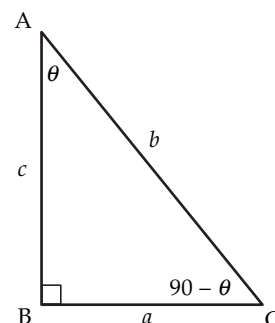
	0°	90°	180°	270°	360°
sin					
cos					
tan					

2

	0°	90°	180°	270°	360°
cosec					
sec					
cot					

If $\angle A = \theta^\circ$ and $\angle B = (90 - \theta)^\circ$ [that is, $\angle A$ and $\angle B$ are complementary] find each of these ratios.

- | | | |
|-----------------------------|-----------------------------|-----------------------------|
| 3 $\sin \theta$ | 4 $\cos \theta$ | 5 $\tan \theta$ |
| 6 $\sin(90 - \theta)^\circ$ | 7 $\cos(90 - \theta)^\circ$ | 8 $\tan(90 - \theta)^\circ$ |



Complete each statement.

- 9 The **sine** of an angle equals the _____ of its complement.
 10 The **cosine** of an angle equals the _____ of its complement.
 11 The **tangent** of an angle equals the _____ of its complement.

Give the value of x .

- | | | | |
|-----------------------------|---|---|-----------------------------|
| 12 $\sin 40^\circ = \cos x$ | 13 $\cos 1^\circ = \sin x$ | 14 $\tan 30^\circ = \cot x$ | 15 $\sin 24^\circ = \cos x$ |
| 16 $\cos 38^\circ = \sin x$ | 17 $\operatorname{cosec} 63^\circ = \sec x$ | 18 $\sec 73^\circ = \operatorname{cosec} x$ | 19 $\tan 47^\circ = \cot x$ |

Give each expression in terms of its complementary angle.

- | | | | |
|-------------------------|---------------------------------------|-------------------------|-------------------------|
| 20 $\sin A$ | 21 $\tan B$ | 22 $\cos C$ | 23 $\cot x$ |
| 24 $\sec \frac{B}{2}$ | 25 $\operatorname{cosec} \frac{A}{2}$ | 26 $\sin 30^\circ$ | 27 $\cos 45^\circ$ |
| 28 $\tan 45^\circ$ | 29 $\cot 50^\circ$ | 30 $\sin(30 + A)^\circ$ | 31 $\cos(60 + A)^\circ$ |
| 32 $\tan(80 + B)^\circ$ | 33 $\sin 36^\circ 45'$ | 34 $\cos 83^\circ 15'$ | 35 $\tan 19^\circ 41'$ |

Evaluate.

- | | | | |
|--------------------------------------|---|---|--|
| 36 $\cos 90^\circ + \sin 0^\circ$ | 37 $\cos 90^\circ \times \sin 90^\circ$ | 38 $\cos 90^\circ \times \sin 90^\circ$ | 39 $\cos^2 90^\circ + \sin^2 90^\circ$ |
| 40 $\cos 270^\circ - \sin 270^\circ$ | 41 $\cot 270^\circ + \cos 180^\circ$ | 42 $\cos 0^\circ + \cos 180^\circ$ | 43 $2 \sin 270^\circ + \sin 90^\circ$ |

29 radian-degree conversions

A **radian** is the angle subtended at the centre of a unit circle (radius of 1 unit) by an arc 1 unit in length.

The circumference of a circle = $2\pi r$

$\therefore$ for a unit circle, $C = 2\pi \times 1 = 2\pi \approx 6.28$

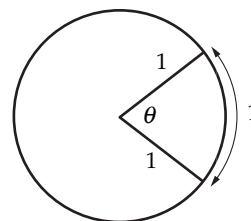
$\therefore$ There are approximately 6.28 radians in a revolution.

$$2\pi \text{ radians} = 360^\circ$$

$$\pi \text{ radians} = 180^\circ$$

$$\therefore 1^\circ = \frac{\pi}{180} \text{ radians}$$

$$\therefore 1 \text{ radian} = \frac{180}{\pi} \text{ degrees}$$



$\theta = 1 \text{ radian}$

e.g. 1 Convert 300° to radians

$$300^\circ = 300 \times \frac{\pi}{180}$$

$$= \frac{5\pi}{3} \text{ radians}$$

or 5.24 radians (to 2 decimal places)

2 Convert $\frac{3\pi}{4}$ radians to degrees.

$$\frac{3\pi}{4} \text{ radians} = \frac{3\pi}{4} \times \frac{180}{\pi} \text{ degrees}$$

$$= 135^\circ$$

Convert to radians, leaving answers in terms of π .

1 45°

2 60°

3 30°

4 40°

5 120°

6 150°

7 300°

8 200°

9 450°

10 225°

11 100°

12 405°

Convert to degrees.

13 $\frac{\pi}{9}$

14 $\frac{\pi}{2}$

15 $\frac{7\pi}{6}$

16 $\frac{4\pi}{3}$

17 $\frac{7\pi}{4}$

18 $\frac{11\pi}{12}$

19 $\frac{5\pi}{4}$

20 $\frac{3\pi}{2}$

21 $\frac{4\pi}{9}$

22 $\frac{7\pi}{18}$

23 $\frac{7\pi}{3}$

24 $\frac{7\pi}{20}$

Note: students should be familiar with all the conversions above.

Convert to radians, leaving answers in terms of π .

25 18°

26 160°

27 70°

28 50°

29 320°

30 340°

31 390°

32 400°

Convert to degrees and minutes.

33 0.5

34 3.1

35 0.8

36 2.1

37 6.1

38 1.3

39 6.5

40 9.3

Convert to radians. Give answers to two decimal places.

41 39°

42 49°

43 121°

44 231°

45 381°

46 380°

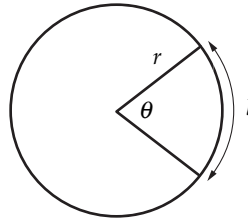
47 400°

48 415°

Length of arc: $l = r\theta$

Area of a sector: $A = \frac{1}{2}r^2\theta$

where r is the radius and θ is the angle subtended at the centre



e.g.

- 1** In a circle of radius 3 cm, find the length of the arc and the area of the sector subtended by an angle of 2.3 radians.

$$l = r\theta$$

$$l = 3 \times 2.3$$

$$= 6.9$$

$$\therefore \text{Arc length } 6.9 \text{ cm}$$

$$A = \frac{1}{2}r^2\theta$$

$$= \frac{1}{2} \times 3^2 \times 2.3$$

$$= 10.35$$

$$\therefore \text{Area } 10.35 \text{ cm}^2$$

- 2** The length of an arc on a circle of radius 5.2 cm is 10.8 cm. Find the angle this arc subtends at the centre of the circle.

$$l = r\theta$$

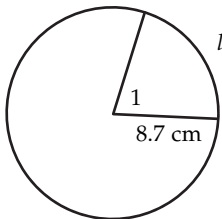
$$10.8 = 5.2 \times \theta$$

$$\therefore \theta = 2.08$$

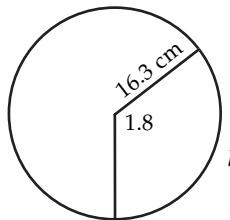
$$\therefore \text{the angle is } 2.08 \text{ radians}$$

Determine the length of each indicated arc, leaving answers in terms of π , where appropriate.

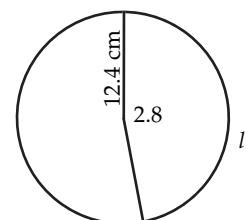
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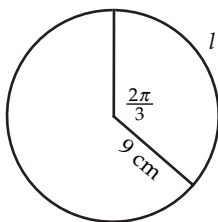
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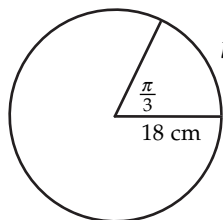
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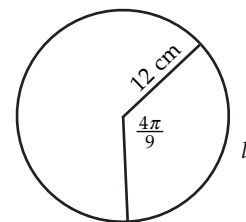
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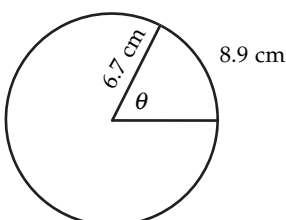
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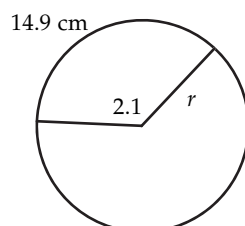
- 7** A sector is removed from a circle of radius 25 cm. The angle this sector makes with the centre of the circle is $\frac{\pi}{5}$ radians. Determine the length of the arc that is left.
- 8** A circle of radius 15 cm has a sector removed. This sector forms an angle of $\frac{3\pi}{5}$ radians at the centre. Determine the perimeter of the remaining sector.

Determine the value of each pronumeral. (Angles are in radians.)

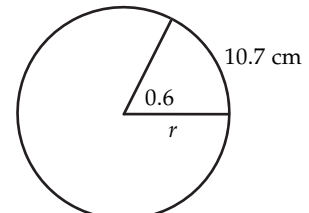
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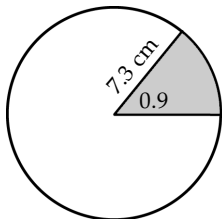
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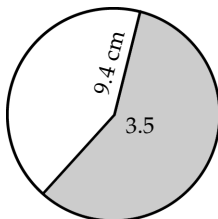
- 12 The arc of a circle is 17.9 cm. Determine the angle subtended by the arc given that the radius is 9.6 cm.

Determine the area of each shaded sector. (Angles are in radians.)

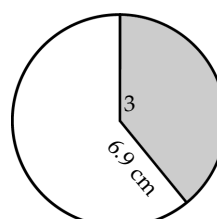
13



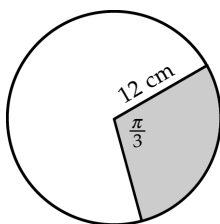
14



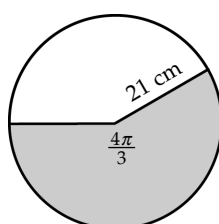
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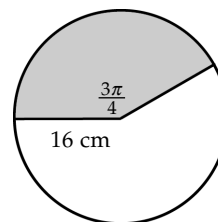
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17

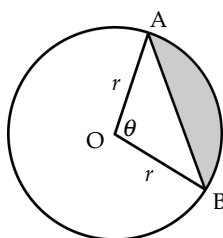


18



- 19 Determine the area of the sector of a circle where the radius is 15 cm and the angle at the centre is $\frac{\pi}{3}$ radians.
- 20 A circle has a radius of 10.8 cm. A sector is removed. This sector forms an angle of 2.6 radians at the centre. Determine the arc length and the area of the sector.
- 21 The area of a sector is 306.5 cm² and the radius of the circle is 12.7 cm. Find the angle (in radians) at the centre of the circle.
- 22 The area of a sector is 377.2 cm². The angle at the centre of the circle is 3.1 radians. Determine the radius.

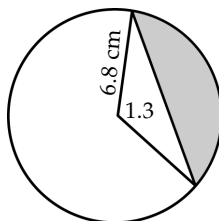
23



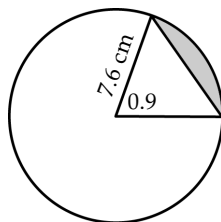
- Write the area of the sector AOB.
- Write the formula that will give the area of $\triangle AOB$.
- Show that the formula for the area of the minor segment AB is given by: $A = \frac{1}{2}r^2(\theta - \sin\theta)$

Determine the area of each shaded segment. (Angles are in radians.)

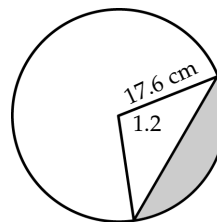
24



25



26



- 27 A segment of a circle is shaded. The radius of the circle is 18.6 cm and the segment subtends an angle of $\frac{3\pi}{4}$ radians at the centre. Determine the area of the **major** segment.

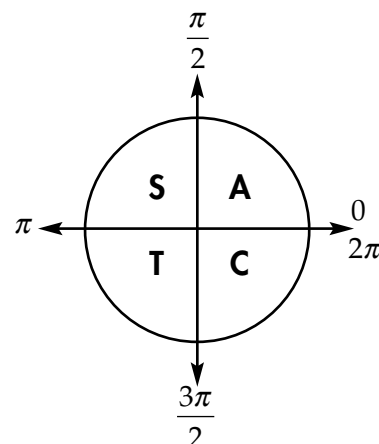
e.g.

- 1** Determine the exact value of $\cos \frac{2\pi}{3}$.

$$\begin{aligned}\cos \frac{2\pi}{3} &= -\cos(\pi - \frac{2\pi}{3}) \\ &= -\cos \frac{\pi}{3} \\ &= -\frac{1}{2}\end{aligned}$$

- 1** Determine the exact value of $\tan \frac{4\pi}{3}$.

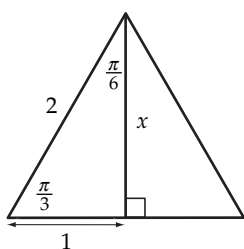
$$\begin{aligned}\tan \frac{4\pi}{3} &= \tan(\pi + \frac{\pi}{3}) \\ &= \tan \frac{\pi}{3} \\ &= \sqrt{3}\end{aligned}$$



Convert each angle to radians by multiplying by $\frac{\pi}{180}$.

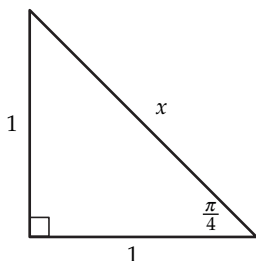
- | | | | |
|---------------------|-----------------------|-----------------------|-----------------------|
| 1 30° | 2 150° | 3 210° | 4 330° |
| 5 60° | 6 120° | 7 240° | 8 300° |
| 9 45° | 10 135° | 11 225° | 12 315° |

- 13** Find the value of x as a surd. Then find each value below.



- | | | |
|---|-------------------------------|-------------------------------|
| a $\sin \frac{\pi}{6}$ | b $\cos \frac{\pi}{6}$ | c $\tan \frac{\pi}{6}$ |
| d $\sin \frac{\pi}{3}$ | e $\cot \frac{\pi}{3}$ | f $\tan \frac{\pi}{3}$ |
| g $\operatorname{cosec} \frac{\pi}{6}$ | h $\sec \frac{\pi}{6}$ | i $\cot \frac{\pi}{6}$ |
| j $\operatorname{cosec} \frac{\pi}{3}$ | k $\sec \frac{\pi}{3}$ | l $\cot \frac{\pi}{3}$ |

- 14** Find the value of x as a surd. Then find each value below.



- | | | |
|---|-------------------------------|-------------------------------|
| a $\sin \frac{\pi}{4}$ | b $\cos \frac{\pi}{4}$ | c $\tan \frac{\pi}{4}$ |
| d $\operatorname{cosec} \frac{\pi}{4}$ | e $\cot \frac{\pi}{4}$ | f $\sec \frac{\pi}{4}$ |

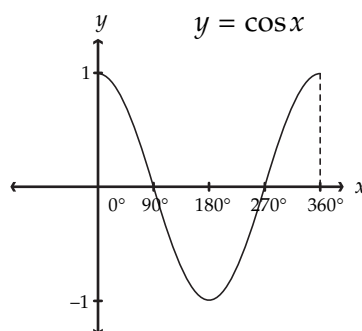
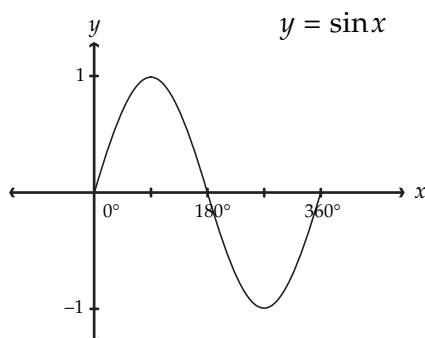
Give the exact value.

- | | | | |
|--|--|---|--|
| 15 $\cos \frac{5\pi}{6}$ | 16 $\tan \frac{3\pi}{4}$ | 17 $\cos \frac{7\pi}{4}$ | 18 $\cos \frac{4\pi}{3}$ |
| 19 $\tan \frac{5\pi}{6}$ | 20 $\sin \frac{\pi}{6}$ | 21 $\sin \frac{5\pi}{6}$ | 22 $\sin \frac{7\pi}{6}$ |
| 23 $\sin \frac{11\pi}{6}$ | 24 $\tan \frac{5\pi}{4}$ | 25 $\tan \frac{7\pi}{4}$ | 26 $\operatorname{cosec} \frac{4\pi}{3}$ |
| 27 $\sin \frac{5\pi}{6} \sin \frac{\pi}{3}$ | 28 $\tan \frac{7\pi}{4} + \cos \frac{2\pi}{3}$ | 29 $\sin^2 \frac{\pi}{3} + \sin \frac{5\pi}{6}$ | 30 $\cos \frac{5\pi}{6} + \sin \frac{2\pi}{3}$ |
| 31 $\cos \frac{3\pi}{4} \times \cos \frac{5\pi}{4}$ | 32 $\sec \frac{5\pi}{6} + \cot \frac{4\pi}{3}$ | 33 $\tan \frac{2\pi}{3} \cdot \cos \frac{5\pi}{6}$ | 34 $\sqrt{2} \cdot \sin \frac{5\pi}{4} \cdot \cos \frac{2\pi}{3}$ |
| 35 $\sin^2 \frac{5\pi}{3} + \cos^2 \frac{5\pi}{3}$ | 36 $\tan^2 \frac{\pi}{3} + \cot^2 \frac{5\pi}{6}$ | 37 $2 \operatorname{cosec} \frac{5\pi}{4} + \sec \frac{7\pi}{4}$ | 38 $\sqrt{3} \cot \frac{\pi}{3} - \sqrt{2} \sec \frac{3\pi}{4}$ |

When sketching $y = \sin x$ or $y = \cos x$ it is only necessary to find the critical or important values. These values occur where the graph cuts the x -axis and y -axis and where it reaches maximum and minimum points.

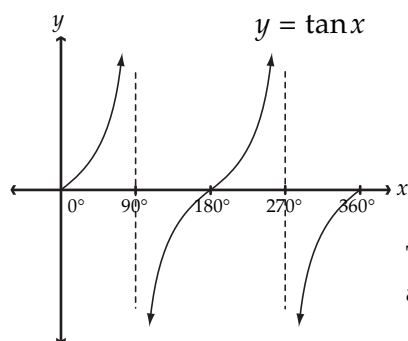
e.g.

- 1** Sketch the graphs of $y = \sin x$ and $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$.



For each graph the amplitude is 1 and the period is 360° .

- 2** Sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$.



The graph is discontinuous at $x = 90^\circ$ and $x = 270^\circ$.

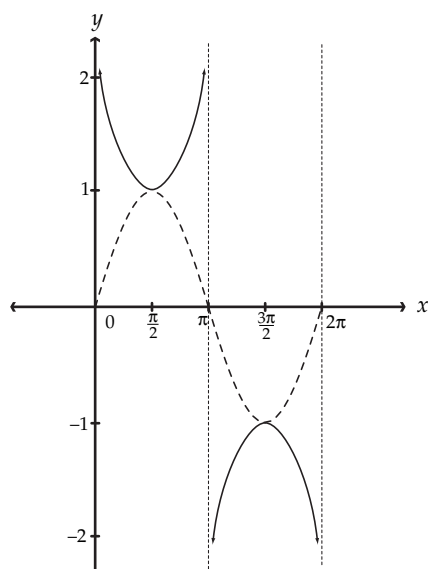
- 1 Give the amplitude of $y = \cos x$.
- 2 Give the period of $y = \sin x$ and $y = \tan x$.
- 3 Sketch the graph $y = \sin x$ for $-180^\circ \leq x \leq 180^\circ$.
From your graph, give approximate values for the following.
 - a $\sin(-45^\circ)$
 - b $\sin 60^\circ$
 - c $\sin 130^\circ$
 - d x , if $\sin x = 0.6$
- 4 Sketch the graph $y = \cos x$ for $-180^\circ \leq x \leq 180^\circ$.
From your graph, give approximate values for the following.
 - a $\cos 45^\circ$
 - b $\cos(-100^\circ)$
 - c $\cos 150^\circ$
 - d x , if $\cos x = 0.6$
- 5 Sketch the graph $y = \tan x$ for $-180^\circ \leq x \leq 180^\circ$.
From your graph, give approximate values for the following.
 - a $\tan 60^\circ$
 - b $\tan 100^\circ$
 - c $\tan(-160^\circ)$
 - d x , if $\tan x = 0.6$

For each of the following, sketch the graph and give the amplitude and the period.

- 6 $y = 1 + \sin x$ for $0^\circ \leq x \leq 360^\circ$
- 7 $y = 2 + \cos x$ for $0^\circ \leq x \leq 360^\circ$
- 8 $y = \cos x - 1$ for $-180^\circ \leq x \leq 180^\circ$
- 9 $y = -\sin x$ for $-180^\circ \leq x \leq 180^\circ$
- 10 $y = -\cos x$ for $0^\circ \leq x \leq 360^\circ$

e.g.

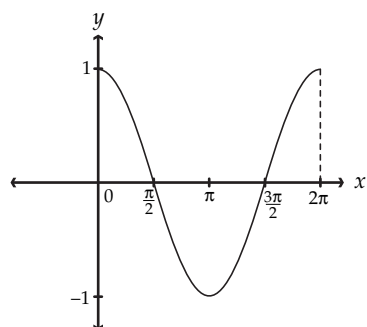
- 1** Sketch the graph of $y = \operatorname{cosec} x$ for $0 \leq x \leq 2\pi$.



Start by drawing the graph of $y = \sin x$. Find the reciprocal of each point on the sine curve to obtain the corresponding value for the cosecant.

When $\sin \theta = 0$, $\operatorname{cosec} \theta$ is undefined. This means that there are discontinuities at π and 2π , or 180° and 360° . That is, the asymptotes for $y = \operatorname{cosec} x$ are at $x = \pi$ ($x = 180^\circ$) and $x = 2\pi$ ($x = 360^\circ$).

- 2** Sketch the graph of $y = 2\cos x$ for $0 \leq x \leq 2\pi$.



The 2 changes the amplitude of the curve but does not affect the period.

- 1 Sketch $y = \cos x$ for $0 \leq x \leq 2\pi$. Then sketch $y = \sec x$ for $0 \leq x \leq 2\pi$ and write the equations of the asymptotes.
- 2 Sketch $y = \tan x$ for $0 \leq x \leq 2\pi$. Then sketch $y = \cot x$ for $0 \leq x \leq 2\pi$ and write the equations of the asymptotes.

Sketch each function in the given domain and give the amplitude and the period of the graph.

3 $y = 2\sin x$ ($0 \leq x \leq 2\pi$)

4 $y = 3\sin x$ ($0^\circ \leq x \leq 360^\circ$)

5 $y = 3\cos x$ ($0^\circ \leq x \leq 360^\circ$)

6 $y = \frac{1}{2}\cos x$ ($0 \leq x \leq 2\pi$)

7 $y = \frac{1}{2}\sin x$ ($-\pi \leq x \leq \pi$)

8 $y = \frac{1}{4}\cos x$ ($-180^\circ \leq x \leq 180^\circ$)

9 $y = 1 + 2\cos x$ ($0 \leq x \leq 2\pi$)

10 $y = 2 + 2\sin x$ ($0^\circ \leq x \leq 360^\circ$)

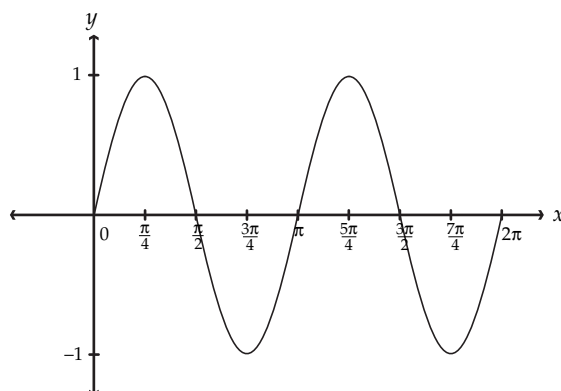
11 $y = 3\cos x - 1$ ($-\pi \leq x \leq \pi$)

12 $y = 2\sin x - 2$ ($0^\circ \leq x \leq 360^\circ$)

For the graph of a function of the type $y = a \cos bx$ the a affects the amplitude of the graph and the b affects the period.

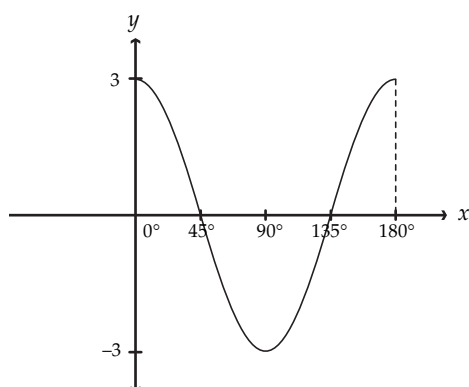
e.g.

- 1** Sketch the graph of $y = \sin 2x$ for $0 \leq x \leq 2\pi$.



The amplitude is 1 and the period is π . That is, the 2 reduces the original period of $y = \sin x$ by a factor of $\frac{1}{2}$; so there are two complete sine curves in the original period of 2π .

- 2** Sketch the graph of $y = 3 \cos 2x$ for $0^\circ \leq x \leq 180^\circ$.



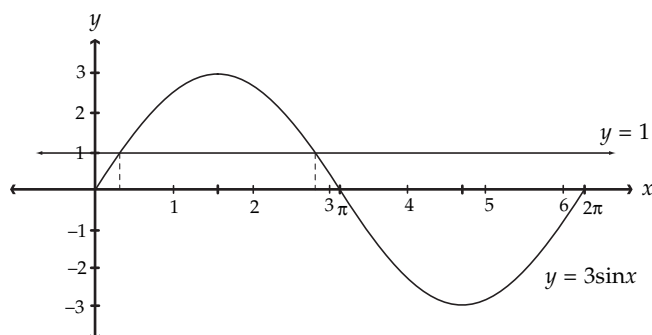
The amplitude is 3, which is 3 times the amplitude of $y = \cos x$. The period is 180° .

Sketch each function in the given domain and give the amplitude and the period.

- | | |
|--|--|
| 1 $y = 3 \sin 2x$ for $0 \leq x \leq 2\pi$ | 2 $y = \frac{1}{2} \cos 2x$ for $0^\circ \leq x \leq 360^\circ$ |
| 3 $y = 2 \sin 3x$ for $0 \leq x \leq \pi$ | 4 $y = 3 \cos 3x$ for $0 \leq x \leq 2\pi$ |
| 5 $y = \frac{1}{2} \cos 4x$ for $0^\circ \leq x \leq 180^\circ$ | 6 $y = 4 \sin 3x$ for $-180^\circ \leq x \leq 180^\circ$ |
| 7 $y = \frac{1}{4} \sin 4x$ for $0 \leq x \leq \pi$ | 8 $y = 1 + 2 \cos 2x$ for $0 \leq x \leq \pi$ |
| 9 $y = 1 + 3 \sin 2x$ for $0^\circ \leq x \leq 180^\circ$ | 10 $y = 1 + 2 \sin 2x$ for $-\pi \leq x \leq \pi$ |
| 11 $y = 1 + 3 \cos 4x$ for $0 \leq x \leq \pi$ | 12 $y = 2 \sin 3x - 2$ for $0^\circ \leq x \leq 180^\circ$ |

e.g.

- 1** Solve graphically the equation $3\sin x = 1$ in the domain $0 \leq x \leq 2\pi$.

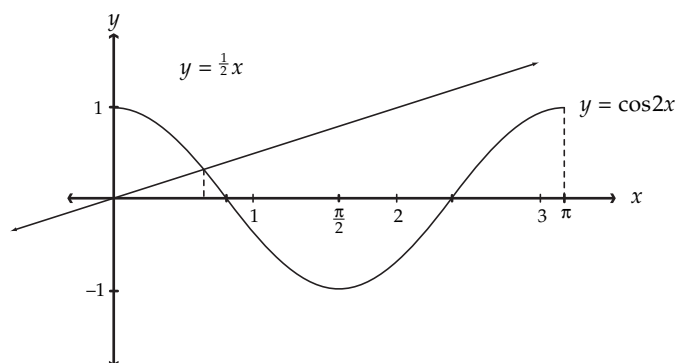


The solution of $3\sin x = 1$ is represented by the intersection of the graphs of $y = 3\sin x$ and $y = 1$. To solve the equation, sketch the two graphs and then take the x -values of the points of intersection.

Remember, the x -axis must be in radians. It is useful if the scale on the y -axis is the same as on the x -axis (π radians ≈ 3.1). The amplitude of $y = 3\sin x$ is 3 and the period is 2π .

The graphical solution of $3\sin x = 1$ is approximately 0.3 and 2.8 radians.

- 2** Solve the equation of $\cos 2x = \frac{1}{2}x$ for $0 \leq x \leq \pi$.



To solve the equation, graph $y = \cos 2x$ and $y = \frac{1}{2}x$. Remember, the amplitude of $y = \cos 2x$ is 1 and the period is π .

The solution of $\cos 2x = \frac{1}{2}x$ is approximately 0.6 radians.

Note that solving equations graphically only gives an approximate answer.

Solve each equation for the given domain by drawing appropriate graphs. Give answers to one decimal place.

- | | |
|--|---|
| 1 $\sin x = \frac{1}{4}$ ($0 \leq x \leq 2\pi$) | 2 $3\cos x = 2$ ($0 \leq x \leq 2\pi$) |
| 3 $\sin 2x = \frac{1}{2}x$ ($-\pi \leq x \leq \pi$) | 4 $3\cos x = x$ ($-\pi \leq x \leq \pi$) |
| 5 $\sin x = \cos x$ ($-\pi \leq x \leq \pi$) | 6 $2\sin 2x = \frac{1}{2}x$ ($0 \leq x \leq \pi$) |
| 7 $\sin x = -\frac{1}{2}$ ($0 \leq x \leq 2\pi$) | 8 $2\cos x = \frac{1}{3}x$ ($0 \leq x \leq 2\pi$) |
| 9 $\cos 2x = -\frac{1}{2}x$ ($-\pi \leq x \leq \pi$) | 10 $\sin 2x = -\frac{1}{4}x$ ($-\pi \leq x \leq \pi$) |
| 11 $2\sin x = \cos x$ ($0 \leq x \leq 2\pi$) | 12 $2\sin 2x = 3\cos 2x$ ($0 \leq x \leq \pi$) |

1 Introduction to Pythagoras' theorem

1 50 2 13 3 12.0 4 9.9 5 10 6 18.0 7 15.8 8 17.0
9 17.5 10 14.4 11 13.0 12 14.4 13 9.7 14 12.6
15 15.2 16 No 17 Yes 18 Yes 19 7.3 m 20 54.8 km

2 Applications of Pythagoras' theorem

1 22.2 cm 2 9.7 m 3 40.1 cm 4 12.5 cm 5a 72.8 cm
5b 66 m 6a 85.3 m 6b 109.2 m 7 7.1 km 8 7.5 cm 9 8.7 m

3 Sine, cosine and tangent ratios: finding sides (whole degrees)

1 30.7 2 12.6 3 2.5 4 5.0 5 5.7 6 19.7 7 11.7 8 31.1
9 19.3 10 41.1 11 5.6 12 22.0 13 16.0 14 643.5 15 23.4
16 19.8 17 11.3 cm 18 15.6 cm 19 41.6 cm 20 36.9 cm
21 22.5 cm

4 Sine, cosine and tangent ratios: finding sides (whole degrees, unknown denominator)

1 37.6 2 30.5 3 9.5 4 20.4 5 40.9 6 37.1 7 38.9
8 78.6 9 49.2 10 12.7 11 21.5 12 9.8 13 50.6
14 50.5 15 40.9 16 5.4 17 18.3 cm 18 14.3 cm
19 21.8 cm 20 16.1 cm 21 6.6 cm

5 Sine, cosine and tangent ratios: finding sides (degrees and minutes)

1 11.9 2 8.0 3 24.7 4 20.3 5 15.0 6 12.4 7 18.8
8 58.9 9 33.4 10 24.6 11 14.8 12 17.9 13 17.9
14 19.6 15 16.6 16 13.0 17 14.9 cm 18 24.6 cm
19 21.3 cm 20 39.1 cm 21 30.7 cm

6 Sine, cosine and tangent ratios: finding sides (degrees and minutes, unknown denominator)

1 8.3 2 20.9 3 22.0 4 15.4 5 6.9 6 21.0 7 4.7
8 18.3 9 48.7 10 16.4 11 13.7 12 14.2 13 8.0
14 18.4 15 19.6 16 20.5 17 24.9 cm 18 44.5 cm
19 5.9 cm 20 40.5 cm 21 21.8 cm

7 Sine, cosine and tangent ratios: finding angles (whole degrees)

1 37° 2 39° 3 35° 4 31° 5 40° 6 44° 7 58° 8 62°
9 30° 10 37° 11 43° 12 52° 13 33° 14 42° 15 51°
16 61° 17 48° 18 55° 19 42° 20 42° 21 48°

8 Sine, cosine and tangent ratios: finding angles (degrees and minutes)

1 $57^\circ 22'$ 2 $37^\circ 30'$ 3 $52^\circ 1'$ 4 $43^\circ 10'$ 5 $31^\circ 26'$ 6 $40^\circ 2'$
7 $52^\circ 12'$ 8 $57^\circ 25'$ 9 $43^\circ 2'$ 10 $34^\circ 31'$ 11 $49^\circ 35'$
12 $46^\circ 51'$ 13 $30^\circ 57'$ 14 $36^\circ 57'$ 15 $46^\circ 1'$ 16 $28^\circ 29'$
17 $46^\circ 47'$ 18 $32^\circ 40'$ 19 $57^\circ 48'$ 20 $26^\circ 44'$ 21 $37^\circ 45'$

9 Sine, cosine and tangent ratios: review (whole degrees)

1 16.8 cm 2 56.2 m' 3 40° 4 24.4 cm 5 44° 6 49°
7 13.6 m 8 62° 9 54° 10 16.8 cm 11 56° 12 7.2 cm
13 58° 14 16.2 m 15 16.8 m 16 41° 17 24.2 cm
18 38° 19 36° 20 89.9 cm 21 40°

10 Sine, cosine and tangent ratios: review (degrees and minutes)

1 $49^\circ 38'$ 2 7.6 cm 3 $32^\circ 12'$ 4 $39^\circ 23'$ 5 9.2 cm
6 $41^\circ 10'$ 7 $45^\circ 54'$ 8 9.0 m 9 49.5 m 10 28.5 cm
11 $42^\circ 29'$ 12 $52^\circ 0'$ 13 $49^\circ 35'$ 14 37.0 cm 15 15.7 cm

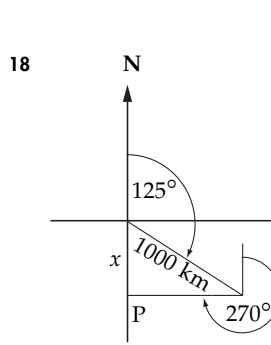
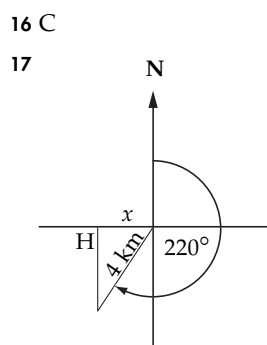
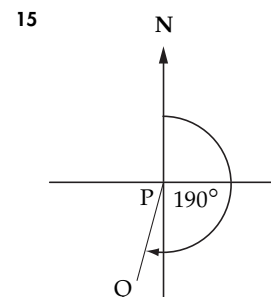
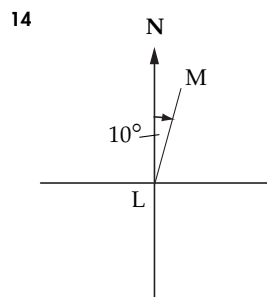
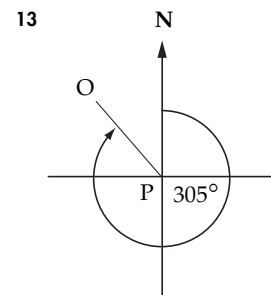
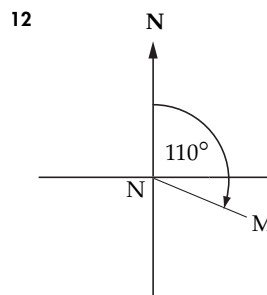
16 15.4 cm 17 39.5 cm 18 $29^\circ 21'$ 19 84.2 cm
20 $48^\circ 04'$ 21 23.9 cm

11 Angles of elevation and depression

1 C, 49 m 2 B, 21 m 3 A, $58^\circ 47'$ 4 B, $70^\circ 32'$ 5 21 m
6 $72^\circ 15'$ 7 162.9 m 8 15 m 9 $38^\circ 53'$ 10 285 m
11 20 m 12 28° 13 17.6 m 14 59.4 m 15 77.7 m
16 11.9 m 17 8 m 18 67° 19 63° 20 7.7 m

12 Introduction to compass bearings

1 A 2 A 3 C 4 B 5 B 6 040° 7 220° 8 340°
9 100° 10 240° 11 345°

**13 Applications of compass bearings**

1 A, 156 km 2 A, 48 km 3 C, 49° 4 B, 6.6 km
5 234° 6 11.5 km 7 475 km 8 027° 9 22.2 km
10 3h 55 mins 11 157° 12 323° 13 219° 14 2231.8 km
15 55° , 29 h 16 608 km 17 33.6 km 18 312°

14 Problems involving trigonometry

1 7 cm 2 53° 3 14.6 m 4 10.5 m 5 25 m 6 78°
7 $47^\circ 10'$ 8 10.70 m 9 $66^\circ 48'$ 10 83° 11 34 cm
12 40 cm 13 Yes 14 10.4 m

15 The sine rule: finding a side (acute angles)

1 8.4 cm 2 11.4 cm 3 11.4 cm 4 31.9 cm 5 8.8 cm
6 11.2 cm 7 16.5 cm 8 26.8 cm 9 $x=56.5$ m, $y=29.9$ m
10 $x=33.6$ m, $y=34.9$ m 11 $x=34.4$ cm, $y=17.2$ cm
12 10.0 cm 13 44.0 m 14 17 km

16 The sine rule: finding an angle (acute angles)

1 $84^\circ 59'$ 2 $44^\circ 7'$ 3 $44^\circ 36'$ 4 $40^\circ 53'$ 5 $57^\circ 58'$
 6 $40^\circ 59'$ 7 $49^\circ 49'$ 8 $59^\circ 10'$ 9 $44^\circ 42'$ 10 $67^\circ 16'$
 11 $48^\circ 58'$ 12 $28^\circ 59'$ 13 $70^\circ 6'$ 14 $51^\circ 50'$ 15 $34^\circ 31'$
 16 $40^\circ 28'$ 17 $61^\circ, 77^\circ$

17 The cosine rule: finding a side (acute angles)

1 12.4 cm 2 14.9 cm 3 9.1 cm 4 9.2 cm 5 10.9 cm
 6 14.3 cm 7 84.9 cm 8 10.6 cm 9 9.2 cm 10 23.3 cm
 11 6.9 cm 12 7.4 cm 13 6.6 cm 14 15.4 cm
 15 22.6 cm 16 13.9 km 17 \$4947.03

18 The cosine rule: finding an angle (acute angles)

1 $84^\circ 46'$ 2 $71^\circ 41'$ 3 $83^\circ 16'$ 4 $49^\circ 59'$ 5 $39^\circ 25'$
 6 $40^\circ 46'$ 7 $64^\circ 46'$ 8 $54^\circ 3'$ 9 Yes 10 $41^\circ 33'$ 11 $85^\circ 25'$
 12 $101^\circ 27', 78^\circ 33'$ 13 $39^\circ 54'$ 14 $80^\circ 1'$ 15 $157^\circ 57'$

19 Area of a triangle

1 95.1 cm² 2 194.2 cm² 3 193.9 m² 4 132.2 cm²
 5 153.0 m² 6 16.7 cm² 7 23.5 cm² 8 18.2 cm²
 9 8.7 cm² 10 131.9 cm² 11 11.9 cm² 12 543.8 m²
 13 166.5 cm² 14 97.1 cm² 15 146.2 cm² 16 8.9 cm
 17 $56^\circ 17'$ 18a 89.22 cm² 18b 68.08 cm² 18c 21.1 cm²
 19 24.4 cm²

20 Angles of any magnitude (degrees)

1 + 2 + 3 – 4 – 5 + 6 – 7 – 8 – 9 – 0.7660
 10 – 0.5446 11 – 0.9325 12 0.1736 13 – 2.1445 14 0.8367
 15 – 0.9397 16 0.3640 17 0.3640 18 0.9397 19 – 0.3640
 20 0.1736 21 50 22 40 23 35 24 80 25 $\cos 34^\circ$
 26 $\sin 40^\circ$ 27 $\tan 40^\circ$ 28 $-\tan 20^\circ$ 29 $-\cos 50^\circ$ 30 $-\sec 30^\circ$
 31 $-\cot 33^\circ$ 32 $\operatorname{cosec} 50^\circ$ 33 $\sin 40^\circ$ 34 $\sin 80^\circ$
 35 $-\cos 60^\circ$ 36 $\tan 40^\circ$ 37 280° 38 260° 39 300°
 40 190° 41 $\cos 30^\circ$ 42 $-\tan 50^\circ$ 43 $-\sin 30^\circ$ 44 $-\sec 40^\circ$

21 Trigonometric equations 1 (degrees)

1 3, 4 2 2, 3 3 1, 2 4 2, 3 5 1, 3 6 1, 3 7 2, 4 8 1, 2
 9 1, 4 10 1, 3 11 3, 4 12 2, 3 13 25° 14 56° 15 56°
 16 40° 17 23° 18 50° 19 44° 20 66° 21 $53^\circ, 307^\circ$
 22 $200^\circ, 340^\circ$ 23 $57^\circ, 237^\circ$ 24 $106^\circ, 286^\circ$ 25 $111^\circ, 249^\circ$
 26 $52^\circ, 128^\circ$ 27 $68^\circ, 292^\circ$ 28 $188^\circ, 352^\circ$ 29 $42^\circ, 222^\circ$
 30 $136^\circ, 224^\circ$ 31 $148^\circ, 328^\circ$ 32 $36^\circ, 144^\circ$ 33 $140^\circ, 220^\circ$
 34 $39^\circ, 141^\circ$ 35 $51^\circ, 231^\circ$ 36 $193^\circ, 347^\circ$ 37 $64^\circ, 296^\circ$
 38 $144^\circ, 324^\circ$ 39 $109^\circ, 251^\circ$ 40 $17^\circ, 163^\circ$ 41 24°
 42 301° 43 326° 44 307° 45 105° 46 198°

22 Sine rule, cosine rule and area of a triangle (acute and obtuse angles)

1 13.6 cm 2 17.7 cm 3 $54^\circ 50'$ 4 $130^\circ 18'$ 5 15.1 cm
 6 $123^\circ 15'$ 7 $133^\circ 0'$ 8 13.2 cm² 9 $88^\circ 11'$ 10 $123^\circ 40'$
 11 $90^\circ 1'$ 12 $127^\circ 36'$ 13 9.9 m 14 88.0 m, 41.3 m
 15 6088.1 cm² 16 $75^\circ 58'$

23 The exact ratios of 30° , 45° and 60°

	30°	60°	45°
sin	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$
tan	$\frac{1}{\sqrt{3}}$	$\sqrt{3}$	1

3 $\sqrt{2}$ 4 $\frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$ 5 $\frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$ 6 1 7 $\frac{\sqrt{3}}{4}$ 8 $\frac{1}{2}$ 9 $\frac{1+\sqrt{3}}{2}$

	30°	60°	45°
cosec	2	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$
sec	$\frac{2}{\sqrt{3}}$	2	$\sqrt{2}$
cot	$\sqrt{3}$	$\frac{1}{\sqrt{3}}$	1

10 1 11 1 12 1 13 $\frac{3}{4}$ 14 1 15 $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$ 16 $\frac{3}{2}$ 17 2
 18 $\frac{1}{3}$ 19 2 20 $\frac{3}{\sqrt{3}} = \sqrt{3}$ 21 4 22 $\frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$ 23 $\frac{\sqrt{3}}{2}$ 24 1
 25 1 26 $2\sqrt{3}$ 27 3 28 $\frac{\sqrt{3}}{4}$ 29 $\frac{1}{2}$ 30 1 31 $1\frac{1}{4}$ 32 $\frac{3}{4}$
 33 $3\sqrt{2}$ 34 $\sqrt{6}$ 35 6

24 Simplifying trigonometric expressions

1 $\cos^2 \theta$ 2 $\cot^2 \theta$ 3 1 4 1 5 $\tan A$ 6 1 7 $\sin \theta$ 8 $\sin^2 \alpha$
 9 $\cot^2 \theta$ 10 $\operatorname{cosec} A$ 11 1 12 $\tan \theta$ 13 1 14 $\operatorname{cosec} \theta$ 15 1
 16 $\cos^2 A$ 17 $\sin^2 A$ 18 3 19 $5\sin^2 \theta$ 20 $4\cos^2 \beta$ 21 $\cos^2 \beta$
 22 $\cos A$ 23 1 24 $2\sin \theta \cos \theta$ 25 $\cos \alpha$
 26 $\operatorname{cosec}^2 \theta + \sec^2 \theta$ 27 2 28 1 29 $\operatorname{cosec}^2 \theta$ 30 $\sin^2 \theta$

26 Trigonometric equations 2

1 $42^\circ 57', 317^\circ 3'$ 2 $47^\circ 19', 132^\circ 41'$ 3 $19^\circ 36', 199^\circ 36'$
 4 $214^\circ 16', 325^\circ 44'$ 5 $111^\circ 58', 248^\circ 2'$ 6 $128^\circ 58', 308^\circ 58'$
 7 $149^\circ 26', 210^\circ 34'$ 8 $99^\circ 1', 279^\circ 1'$ 9 $45^\circ 38', 314^\circ 22'$
 10 $22^\circ 12', 202^\circ 12'$ 11 $219^\circ 21', 320^\circ 39'$ 12 $60^\circ, 300^\circ$
 13 $30^\circ, 210^\circ$ 14 $225^\circ, 315^\circ$ 15 $30^\circ, 330^\circ$ 16 $75^\circ 31', 284^\circ 29'$
 17 $45^\circ, 135^\circ, 225^\circ, 315^\circ$ 18 $19^\circ 28', 160^\circ 32'$
 19 $194^\circ 29', 345^\circ 31'$ 20 $60^\circ, 120^\circ$ 21 $120^\circ, 240^\circ$ 22 $150^\circ, 330^\circ$
 23 $17^\circ 35', 197^\circ 35'$ 24 $41^\circ 49', 138^\circ 11'$ 25 $116^\circ 34', 135^\circ, 296^\circ 34', 315^\circ$
 26 $45^\circ, 225^\circ, 116^\circ 34', 296^\circ 34'$
 27 $41^\circ 49', 138^\circ 11', 210^\circ, 330^\circ$ 28 $60^\circ, 300^\circ$ 29 $90^\circ, 221^\circ 49', 318^\circ 11'$
 30 $15^\circ, 75^\circ, 195^\circ, 255^\circ$ 31 $30^\circ, 120^\circ, 210^\circ, 300^\circ$ 32 $63^\circ 26', 116^\circ 34', 243^\circ 26', 296^\circ 34'$
 33 $113^\circ 29', 156^\circ 31', 293^\circ 29', 336^\circ 31'$ 34 $20^\circ, 40^\circ, 140^\circ, 160^\circ, 260^\circ, 280^\circ$ 35 90° 36 $60^\circ, 300^\circ$

27 Ratios involving surds and angles of any size

1 $\frac{3}{5}, \frac{4}{5}$ 2 $\frac{\sqrt{7}}{4}, \frac{3}{\sqrt{7}}$ 3 $\frac{\sqrt{35}}{6}, \sqrt{35}$ 4 $\frac{7}{\sqrt{65}}, \frac{\sqrt{65}}{4}$ 5 $\frac{\sqrt{45}}{7}, \frac{2}{\sqrt{45}}$
 6 $\frac{6}{\sqrt{85}}, \frac{\sqrt{85}}{7}$ 7 $\frac{\sqrt{2}}{\sqrt{11}}, \frac{3}{\sqrt{2}}$ 8 $\frac{3}{2}, \frac{\sqrt{13}}{3}$ 9 $\frac{3}{\sqrt{11}}, \frac{\sqrt{11}}{2}$ 10 $\frac{\sqrt{22}}{5}, \frac{\sqrt{22}}{\sqrt{3}}$
 11 $\sqrt{1-t^2}, \frac{\sqrt{1-t^2}}{t}$ 12 $\frac{\sqrt{t^2-1}}{t}, \sqrt{t^2-1}$ 16 $\frac{4}{\sqrt{13}}, -\frac{\sqrt{3}}{\sqrt{13}}$
 17 $\frac{7}{\sqrt{40}}, \frac{3}{\sqrt{40}}$ 18 $-\frac{3}{\sqrt{40}}, \frac{\sqrt{40}}{7}$ 19 $-\frac{6}{\sqrt{61}}, -\frac{\sqrt{61}}{5}$ 20 $\frac{\sqrt{7}}{3}, -\frac{4}{\sqrt{7}}$
 21 $\frac{5}{\sqrt{32}}, \frac{\sqrt{32}}{\sqrt{7}}$

28 Ratios of 0° , 90° , 180° , 270° , 360° and complementary angles

1	0	1	0	-1	0
	1	0	-1	0	1
	0	-	0	-	0

2	-	1	-	-1	-
	1	-	-1	-	1
	-	0	-	0	-

3 $\frac{a}{b}$ 4 $\frac{a}{b}$ 5 $\frac{a}{c}$ 6 $\frac{c}{b}$ 7 $\frac{a}{b}$ 8 $\frac{a}{b}$ 9 cosine 10 sine
 11 cotangent 12 50° 13 89° 14 60° 15 66° 16 52°
 17 27° 18 17° 19 43° 20 $\cos(90-A)$ 21 $\cot(90-B)$
 22 $\sin(90-C)$ 23 $\tan(90-x)$ 24 $\operatorname{cosec}(90-\frac{B}{2})$
 25 $\sec(90-\frac{A}{2})$ 26 $\cos 60^\circ$ 27 $\sin 45^\circ$ 28 $\cot 45^\circ$
 29 $\tan 40^\circ$ 30 $\cos(60-A)^\circ$ 31 $\sin(30-A)^\circ$ 32 $\cot(10-B)^\circ$
 33 $\cos 53^\circ 15'$ 34 $\sin 6^\circ 45'$ 35 $\cot 70^\circ 19'$ 36 0 37 0
 38 0 39 1 40 1 41 -1 42 0 43 -1

29 Radian-degree conversions

1 $\frac{\pi}{4}$ 2 $\frac{\pi}{3}$ 3 $\frac{\pi}{6}$ 4 $\frac{2\pi}{9}$ 5 $\frac{2\pi}{3}$ 6 $\frac{5\pi}{6}$ 7 $\frac{5\pi}{3}$ 8 $\frac{10\pi}{9}$ 9 $\frac{5\pi}{2}$ 10 $\frac{5\pi}{4}$
 11 $\frac{5\pi}{9}$ 12 $\frac{9\pi}{4}$ 13 20° 14 90° 15 210° 16 240° 17 315°
 18 165° 19 225° 20 270° 21 80° 22 70° 23 420° 24 63°
 25 $\frac{\pi}{10}$ 26 $\frac{8\pi}{9}$ 27 $\frac{7\pi}{18}$ 28 $\frac{5\pi}{18}$ 29 $\frac{16\pi}{9}$ 30 $\frac{17\pi}{9}$ 31 $\frac{13\pi}{6}$ 32 $\frac{20\pi}{9}$

33 $28^{\circ}39'$ 34 $177^{\circ}37'$ 35 $45^{\circ}50'$ 36 $120^{\circ}19'$ 37 $349^{\circ}30'$
 38 $74^{\circ}29'$ 39 $372^{\circ}25'$ 40 $532^{\circ}51'$ 41 0.68 42 0.86
 43 2.11 44 4.03 45 6.65 46 6.63 47 6.98 48 7.24

30 Length of an arc and area of a sector (radians)

1 8.7 cm 2 29.34 cm 3 34.72 cm 4 6π cm 5 6π cm
 6 $\frac{16\pi}{3}$ cm 7 45π cm 8 $(30+21\pi)$ cm 9 1.3 rads
 10 7.1 cm 11 17.8 cm 12 1.9 rads 13 24.0 cm
 14 154.6 cm^2 15 71.4 cm^2 16 $24\pi \text{ cm}^2$ 17 $294\pi \text{ cm}^2$
 18 $96\pi \text{ cm}^2$ 19 $\frac{75}{2}\pi$ 20 $l=28.1$ cm, $A=151.6 \text{ cm}^2$
 21 3.8 rads 22 15.6 cm 23 a $\frac{1}{2}r^2\theta$ b $\frac{1}{2}ab\sin\theta$
 24 7.8 cm^2 25 3.4 cm^2 26 41.5 cm^2 27 801.6 cm^2

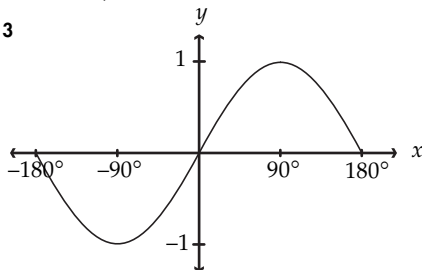
31 Exact values of trigonometric ratios ($\frac{\pi}{3}, \frac{\pi}{4}, \frac{\pi}{6}$ and related angles of any value)

1 $\frac{\pi}{6}$ 2 $\frac{5\pi}{6}$ 3 $\frac{7\pi}{6}$ 4 $\frac{11\pi}{6}$ 5 $\frac{\pi}{3}$ 6 $\frac{2\pi}{3}$ 7 $\frac{4\pi}{3}$ 8 $\frac{5\pi}{3}$ 9 $\frac{\pi}{4}$ 10 $\frac{3\pi}{4}$
 11 $\frac{5\pi}{4}$ 12 $\frac{7\pi}{4}$ 13 $\sqrt{3}$ a $\frac{1}{2}$ b $\frac{\sqrt{3}}{2}$ c $\frac{1}{\sqrt{3}}$ d $\frac{\sqrt{3}}{2}$ e $\frac{1}{2}$ f $\sqrt{3}$ g 2
 h $\frac{2}{\sqrt{3}}$ i $\sqrt{3}$ j $\frac{2}{\sqrt{3}}$ k 2 l $\frac{1}{\sqrt{3}}$ 14 $\sqrt{2}$ a $\frac{1}{\sqrt{2}}$ b $\frac{1}{\sqrt{2}}$ c 1 d $\sqrt{2}$
 e 1 f $\sqrt{2}$ 15 $-\frac{\sqrt{3}}{2}$ 16 -1 17 $\frac{1}{\sqrt{2}}$ 18 $-\frac{1}{2}$ 19 $-\frac{1}{\sqrt{3}}$ 20 $\frac{1}{2}$
 21 $\frac{1}{2}$ 22 $-\frac{1}{2}$ 23 $-\frac{1}{2}$ 24 1 25 -1 26 $-\frac{2}{\sqrt{3}}$ 27 $\frac{\sqrt{3}}{4}$ 28 $-\frac{3}{2}$
 29 $\frac{5}{4}$ 30 0 31 $\frac{1}{2}$ 32 $-\frac{1}{\sqrt{3}}$ 33 $\frac{3}{2}$ 34 $\frac{1}{2}$ 35 1 36 6 37 $-\sqrt{2}$
 38 3

32 Graphs of $\sin\theta$, $\cos\theta$ and $\tan\theta$

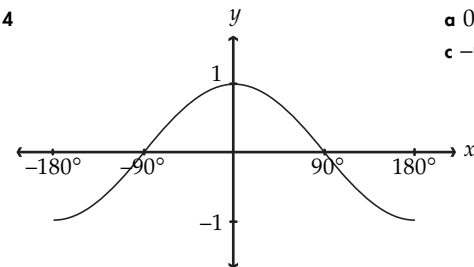
1 1 2 $2\pi, \pi$

3



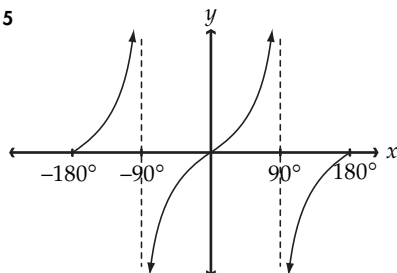
a -0.7 b 0.9
c 0.8 d 37°

4



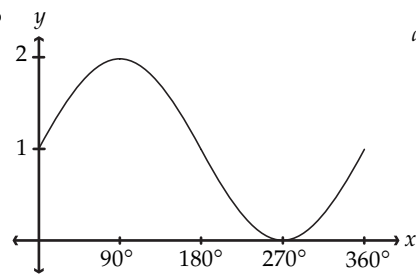
a 0.7 b -0.2
c -0.9 d $\pm 53^{\circ}$

5



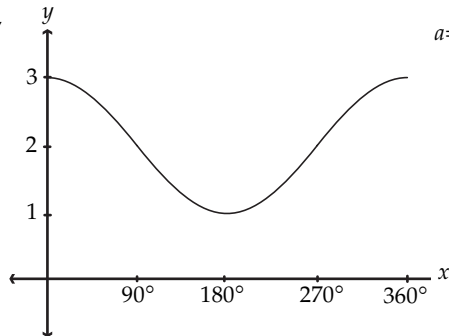
a 1.7 b -5.7 c 0.36
d $-149^{\circ}, 31^{\circ}$

6



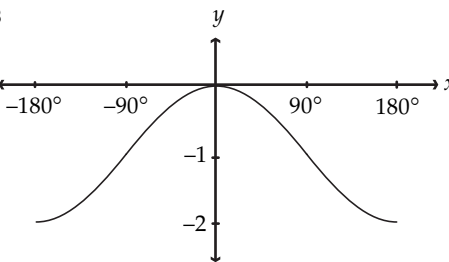
$a=1, P=360^{\circ}$

7



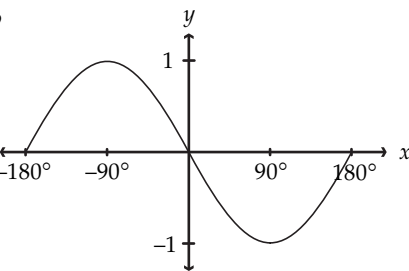
$a=1, P=360^{\circ}$

8



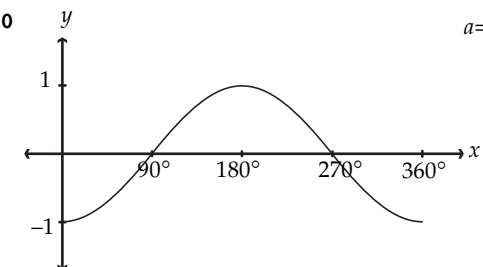
$a=2, P=360^{\circ}$

9



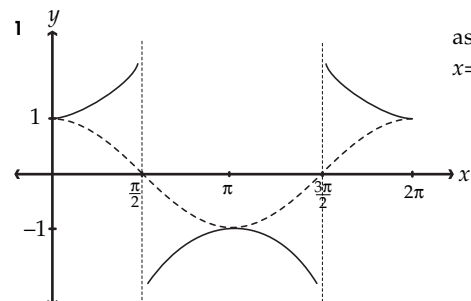
$a=1, P=360^{\circ}$

10

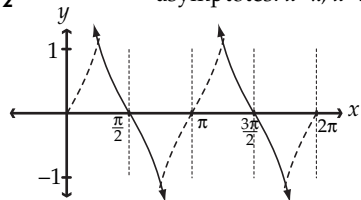
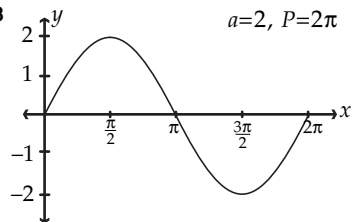
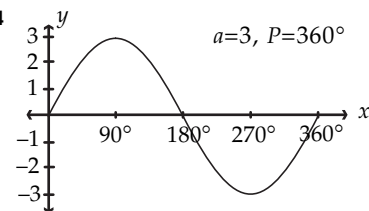
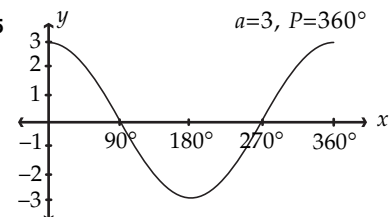
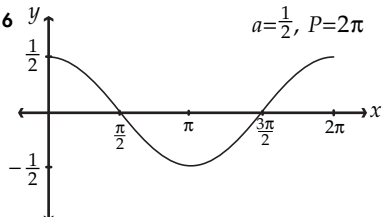
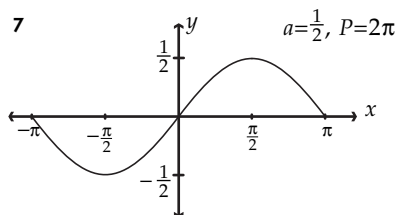
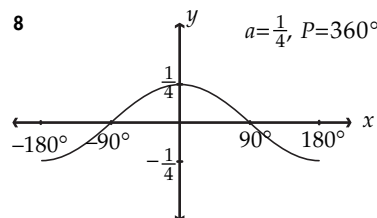
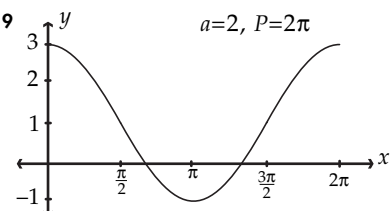
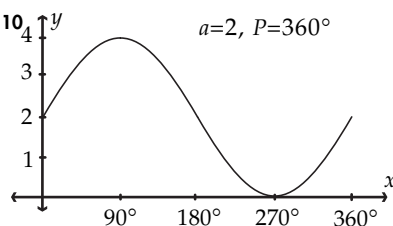
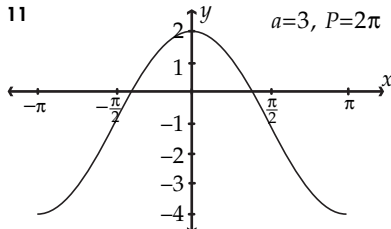
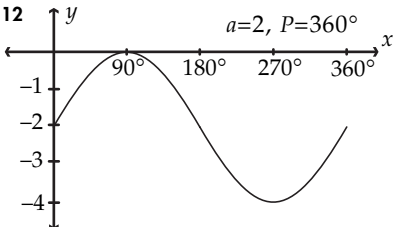
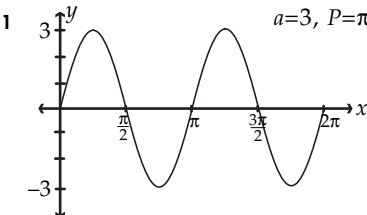
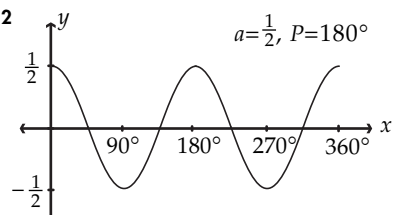
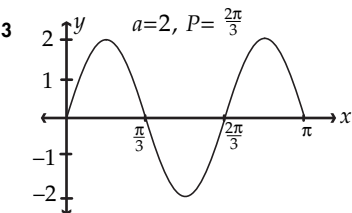
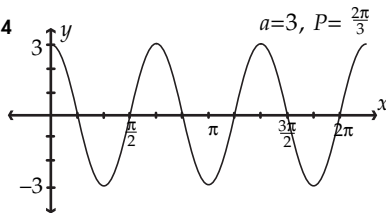
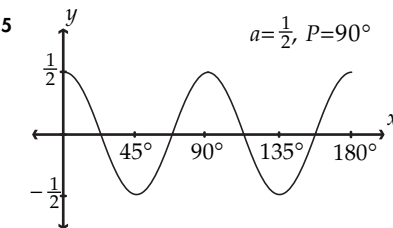
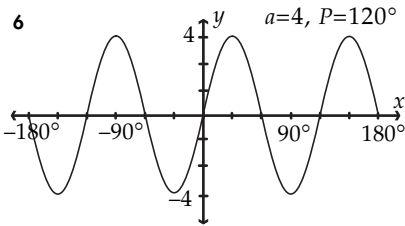
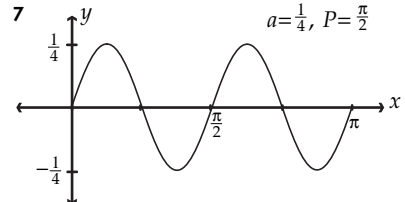
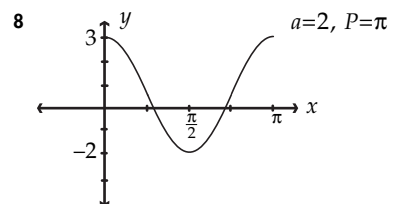
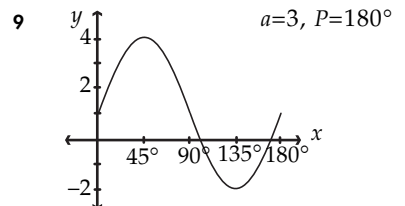
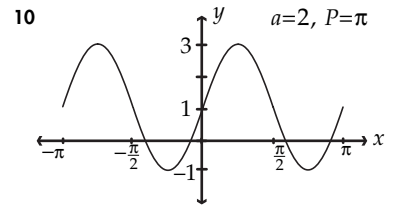
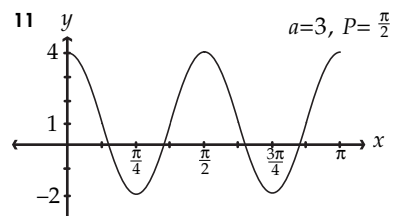
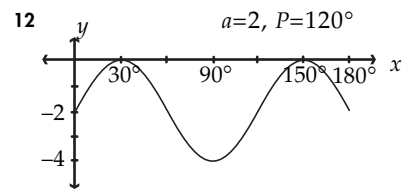


$a=1, P=360^{\circ}$

33 Graphs of reciprocal ratios, $y=\frac{1}{\sin x}$, $y=\frac{1}{\cos x}$



asymptotes:
 $x=\frac{\pi}{2}, x=\frac{3\pi}{2}$

2 asymptotes: $x=\pi$, $x=2\pi$, $x=0$ 3 $a=2$, $P=2\pi$ 4 $a=3$, $P=360^\circ$ 5 $a=3$, $P=360^\circ$ 6 $a=\frac{1}{2}$, $P=2\pi$ 7 $a=\frac{1}{2}$, $P=2\pi$ 8 $a=\frac{1}{4}$, $P=360^\circ$ 9 $a=2$, $P=2\pi$ 10 $a=2$, $P=360^\circ$ 11 $a=3$, $P=2\pi$ 12 $a=2$, $P=360^\circ$ **34 Graphs of $y=a\sin bx$ and $y=a\cos bx$** 1 $a=3$, $P=\pi$ 2 $a=\frac{1}{2}$, $P=180^\circ$ 3 $a=2$, $P=\frac{2\pi}{3}$ 4 $a=3$, $P=\frac{2\pi}{3}$ 5 $a=\frac{1}{2}$, $P=90^\circ$ 6 $a=4$, $P=120^\circ$ 7 $a=\frac{1}{4}$, $P=\frac{\pi}{2}$ 8 $a=2$, $P=\pi$ 9 $a=3$, $P=180^\circ$ 10 $a=2$, $P=\pi$ 11 $a=3$, $P=\frac{\pi}{2}$ 12 $a=2$, $P=120^\circ$ **35 Using graphs to solve trigonometric equations**

- 1 2.9, 0.3 2 0.8, 5.4 3 0, ± 1.2
 4 -2.7, 1.2 5 0.8, -2.4 6 0, 1.4
 7 3.7, 5.8 8 1.3 9 -0.6, 1.1, 1.8
 10 ± 1.8 , 0, ± 2.8 11 0.5, 3.6
 12 0.5, 2.1

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